Economic Issue Prediction Using Partial Least Squares (PLS) & Artificial Neural Network (ANN) Hybrid Model

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A thesis submitted in partial fulfillment of the requirement for the degree of Bachelor of Science in Software Engineering

Summer 2026

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APPROVAL

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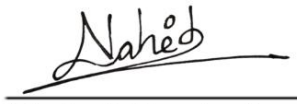
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ACKNOWLEDGEMENTS

First and foremost, I would like to express my deepest gratitude to my supervisor, Dr. Imran Mahmud, Professor & Head of the Department of Software Engineering at Daffodil International University, for his invaluable guidance, continuous support, and immense patience throughout this research. His profound knowledge and insightful feedback were instrumental in shaping this thesis. I also extend my sincere thanks to the faculty members of the Department of Software Engineering (Major in Data Science) at Daffodil International University for providing a solid academic foundation and a conducive learning environment. Finally, I am profoundly grateful to my family and friends for their unwavering encouragement, emotional support, and understanding during the challenging phases of this research. This achievement would not have been possible without them.

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Dedication

This thesis belongs to the quest for knowledge, innovation and excellence in the areas of Software Engineering, Machine Learning, Artificial Intelligence, and Data Science.

It is also devoted to the scientists, academicians, teachers and students who have always been striving and making tremendous efforts in the field of scientific research to promote the knowledge of the world. This research has been inspired by the impact of their work in the areas of computational intelligence, predictive analytics and intelligent decision-making systems.

The work is also dedicated to the broader education community involved in multi-disciplinary research and technology development of complex problems to address real world challenges. My hope is that this book, in a small way, will help in the continuing evolution of hybrid machine learning approaches to economic forecasting and will be a helpful resource for future researchers on the development of intelligent predictive systems.

Lastly, this thesis is dedicated to the ideals of curiosity, critical thinking and lifelong learning which continue to motivate scientific discovery and technological progress.

I would like to thank my respected supervisor, Dr. Imran Mahmud for his expert guidance and immense support in conducting this research as well as for his constructive remarks, continuous encouragement and valuable guidance. His knowledge and support have been extremely useful in the successful completion of this thesis.

Lastly, I would like to acknowledge the contributions of all the teachers, researchers, and students who strive to advance the field of software engineering, machine learning, artificial intelligence and data science. It is my hope that this research will contribute, in some small way, to the advancement of intelligent economic forecasting systems and that this research will inspire some innovation in this field.

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Abstract

In an increasingly interconnected, complex and dynamic global financial system, the accurate prediction of economic crises is increasingly important. Especially in the economics field, high-dimensional and nonlinear data may be beyond the scope of traditional statistical methods. Considering these challenges, this paper proposes a hybrid prediction model with PLS regression and ANN technology, so as to enhance the prediction performance efficiency. The primary objective achieved here is to develop a stable and accurate model, the model is able to detect the nonlinear relationships between the economic data and provide high accuracy forecasts. Machine learning and advanced statistical techniques were applied to preprocess and analyze the data to obtain multidimensional financial/economic variables. The PLS results were good for dimensionality reduction and extraction of latent features, with the optimal selected components being 24, which account for the dataset's maximum variance. The extracted elements were then served as input to the ANN model to enhance the prediction capability and to remove noise and multicollinearity. Identification of the best model: Cross validation results of average RMSE, MAE and R2 Score of 9 models for the proposed Hybrid PLS-ANN model. The results of the experiments demonstrated that the hybrid model's performance is significantly better than the performance of various baseline traditional and ML models. Proposed PLS-ANN model had very good predictive accuracy and strong generalization performance with an R^2 score of 0.9754, RMSE of 1091.56, MAE of 813.30 The comparative analysis results showed that the stand-alone ANN (R^2 = 0.7500), Linear Regression (R^2 = 0.8843), AdaBoost (R^2 = 0.9233), Ridge Regression (R^2 = 0.8531), Lasso Regression (R^2 = 0,8791) and Elastic Net regression (R^2= 0,8414) were inferior to the combined approach presented here findings→ The comparative analysis. Results indicate that PLS-ANN can extract linear and non-linear patterns contained in economic data and provide stable and correct estimation of future values. It is a discovery that brings a new level of scalability and computational efficiency to the economics of forecasting by providing a hybrid modeling approach to aid decision-making by policy makers, researchers, and financial analysts. Moreover, the study shows the ability of hybrid ML approaches in addressing complex economic prediction problems and can serve as a benchmark for future works related to the intelligent systems for econometric forecasting.

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CHAPTER 1- INTRODUCTION

1.1 Background of Study

When it comes to forecasting of economic indicators (GDP growth, financial indicators, market demand etc.), it has a critical importance for policymakers and business decision-makers. Economic data is typically multidimensional and has a lot of dependencies. Traditional linear models of the one-way analysis of variance (ANOVA), like ordinary least squares (OLS) or ARIMA, are not reliable for the nonlinear patterns or many correlated predictors. Modern forecasts have to deal with such complexities, hence the growing use of machine learning (ML). One of the most popular applications of artificial neural networks (ANNs) in finance/economics is as a very powerful non-linear function approximator. In a similar fashion, Partial Least Squares (PLS) regression is a dimensionality-reduction technique which, even in the presence of multicollinearity, is able to project variables onto latent factors. The proposed method involves PLS which filters out the most useful components followed by an ANN that model the nonlinearities (Saha 2021). Therefore, a hybrid PLS-ANN can seemingly take into account at least two benefits: dimensionality reduction and model non-linearity. Interestingly, hybrid PLS-ANN has shown to be more accurate in predicting consumer preference and other fields. The aim of this thesis is to analyze a hybrid framework for predicting economic problems based on a financial data database, where it will have been better to predict the economic problems compared to the conventional statistical methods.

1.2 Problem Statement

A common problem for economists and analysts is to predict economic trends using large and noisy data. Explain the problems of conventional approaches – undershooting (linear systems where non-linear effects are ignored) or overfitting (complex structures on high dimensional data). Our challenge is to build a predictive model of an economic variable – such as the trend in profit or sales – that reflects complexity, both in terms of dimensionality reduction and nonlinear patterns. We examine if the PLS-ANN hybrid model is able to improve the performance of each individual model in predicting profit based on several economic data.

1.3 Research Objectives

The main goals of the current study are: Designing a hybrid PLS-ANN predictive model, conducting PLS and subsequently using the latent factors in an ANN regressor. Apply the model to some observation in economics. Use the financial data set given by user for training and testing of the model (Categorical factors: Product, Country). **Compare performance.** Compare hybrid model to baseline models, which are linear regression, regularized regressions, AdaBoost and ANNs without PLS. **Analyze results.** Understand and present the model results, plot forecasts and evaluate the ability of the model to explain economic trends.

1.4 Research Questions

Some of the key research questions are:

- By contrast to conventional models, how is the capability of the PLS-ANN hybrid in forecasting profit?
- Which latent features do PLS choose as the most important for the target economic variable?
- So, what is the optimum ANN architecture and hyperparameters to accomplish this?
- Can the hybrid model generalize (avoid overfitting) to unseen test data?
- Can a hybrid PLS-ANN model effectively predict economic issues using financial and economic indicators?
- Does Partial Least Squares feature extraction improve the predictive performance of Artificial Neural Networks?
- How does the proposed PLS-ANN model compare with ANN, Linear Regression, Ridge Regression, Lasso Regression, Elastic Net Regression, and AdaBoost Regression?
- What is the optimal number of latent components required to maximize prediction accuracy?
- Which financial and economic factors contribute most significantly to economic issue prediction?
- Can the proposed hybrid framework provide a robust and reliable approach for economic forecasting and decision-making?
- What nonlinear relationships among financial indicators can be captured by ANN but overlooked by traditional statistical models?

- How does dimensionality reduction through PLS affect computational efficiency and model complexity?
- Can the proposed PLS-ANN framework serve as a reliable tool for assisting policymakers, business analysts, and financial decision-makers in economic forecasting?

1.5 Scope of the Study

Nature of Time Series Data. This paper deals with a supervised regression task for static financial data. Target Variable: Profit from sales. Input features:

1. Sales volumes, prices, costs

2. Categorical labels (market segment, country, product type) and quantitatively grouped values (discount bands).

All the standard preprocessing (imputation, encoding, normalization) steps are used and only comparison is made with commonly used regression algorithms. Time series forecasting techniques are ignored as the data are not time-series but rather cross-sectional/date tagged. The study is not directly concerned with the implications for macroeconomic policy but rather is a methodological exercise in forecasting.

1.6 Significance of the study

The book is a helpful demonstration of a viable predictive model, and a valuable addition to the area of computational economics. Having a good prediction about Financial Outcomes could help businesses and policymakers predict trends. The model is developed by combining PLS and ANN, which solves two commonly found problems (dimensionality/multicollinearity & nonlinearity). So, a hybrid solution can become more valuable in all the areas that have abundant and complex data such as Financial Analysis, Risk management, Resource Planning, to mention a few. The methodology is well described in such a way that it is reproducible and can be used by others to implement or add.

1.7 Research Motivation

The driving force is twofold: first, more and more financial data is available and second, with such massive amounts of data, good predictive tools are also needed. Real-time intelligence is moving towards data-driven decision making and advanced analytics (machine learning) is a critical component. Our past research reveals that machine learning methods tend to outperform conventional econometric models in stable times. We sought to be able to use them and test a new hybrid model on true data from the capital markets.

1.8 Thesis Structure

In Chapter 2, we will have a literature review of existing literature on economic forecasting (as a whole) and machine learning in finance research with chapter on PLS, ANN and hybrid models. Chapter 3 shows the methodology of our research, including the datasets, the data preprocessing techniques we used, and the design of our models and the evaluation metrics. Experimental Results, Model Comparisons, and Discussion of Findings. Chapter 5 wraps up the thesis with a summary of the key results, limitations, and suggestions for future research.

CHAPTER 2 – Literature Review

2.1 Introduction

In risk management, policy-making, investing and organization sustainability, predicting the state of economic and financial distress is an increasingly important research area. Logistic Regression, Multiple Linear Regression and Discriminant Analysis are classic statistical techniques which have been extensively employed to forecast financial and economic fragility. The methods are usually not able to capture complex nonlinear relationships in economic data. With the improvement of the prediction accuracy and the robustness of the predictions, academics have increasingly turned to the use of Artificial Intelligence (AI), Machine Learning (ML) and hybrid modelling methods.

ANNs are one of the new methods that have proven to have high capacity for modelling nonlinear relationships. Multicollinearity and high-dimensional data are dealt with by means of dimensionality reduction methods like Partial Least Squares (PLS) and Principal Component Analysis (PCA). In order to enhance the prediction performance, hybrid models based on the combination of feature extraction methods and machine learning algorithms have been discovered to be promising. In this chapter new and relevant financial distress prediction studies, hybrid machine learning systems, neural networks, and dimensionality reduction techniques are discussed. The literature review gives the theoretical and methodological foundations for the proposed hybrid model of Partial Least Squares - Artificial Neural Network (PLS-ANN) for prediction of economic issues..

2.2 Review of Related Literature

2.2.1 FINNET: A Hybrid Deep Learning Network Analysis and Ensemble Learning Model for Financial Distress Prediction

Kumar et al. (2025) suggest a hybrid approach for predicting financial distress, FINNET, which combines Artificial Neural Networks (ANN) with network analysis and ensemble learning. The system consisted of a combination of all above K-best feature, K-means clustering, Voting Classifiers and Stacking Classifiers in a single system. Non-linear learning was done using ANN models and the ensemble methods to enhance the stability and robustness of the classification. The researchers decided to use Python technologies, such as Flask, Scikit-learn and NumPy, and a MySQL database environment to implement the framework. This was found to be significant improvement in prediction accuracy compared to existing machine learning techniques. It shows the importance and rise of deep learning and ensemble systems in the field of predictive analytics. A combination of learning strategies resulted in a high predictive accuracy of FINNET and a robust model.

Research Gap: Feature extraction using Partial Least Squares was not investigated, and may be a tool that could be used to further improve model efficiency and interpretation.

2.2.2 A Hybrid Network Analysis and Machine Learning Model for Enhanced Financial Distress Prediction

Sruneethi and Vinodhkumar (2025) proposed an integrated approach based on network analytics and machine learning techniques to overcome these problems for better prediction of financial distress. For this purpose, Financial Distress dataset was used from Kaggle and financial variables were related to network-based relationships before applying machine learning classification algorithms.

The methodology proposed involves the use of ensemble learning technique, namely Random Forest and Voting Classifier algorithms in the classification of the financially distressed organizations. The data set was divided into two groups: the training set and the test set, to test the performance of the model. For capturing interactions among variables, network analysis was used, which allowed for the discovery of hidden patterns and dependencies.

Results demonstrated that it was statistically significantly better when the network analysis approach was used in applications than the individual machine learning algorithms. The results indicated that the incorporation of structural relationships and the machine learning algorithms can improve the

accuracy of the classification and the interpretation of the model.

Research Gap: This research makes some contribution to the growing research on hybrid financial projections as financial practice. It demonstrates that auxiliary information, beyond what is commonly employed by machine learning classification, can greatly help improve the forecasting accuracy.

Limitation- There was no detailed analysis of dimensionality problems, nor was any analysis of latent variable extraction techniques done with the framework, such as Partial Least Squares.

2.2.3 Hybrid Model for Financial Distress Prediction: Evidence from Vietnamese Listed Firms

To predict financial distress of Vietnamese listed companies, Nguyen D. T et al. (2024) suggested a hybrid machine learning model using the Altman Z-score model and multilayer perceptron artificial neural network (MLP-ANN). Our sample is a panel data set of 509 firms from 2006 to 2019. It is unbalanced panel data of 7,126 observations using financial data.
First, the researchers relied on the Altman Z-score model to choose financial indicators that are relevant to corporate distress. These indicators were used as input to a multilayer neural network architecture that was used to classify firms into distressed or non-distressed. The artificial neural network (ANN) model was developed using past financial ratios and iteratively optimized. The overall prediction accuracy was 98.4%, which was significantly higher than a number of traditional statistical approaches. The findings highlighted the importance of combining domain-specific financial metrics and nonlinear learning processes. Moreover, the research pointed out that hybrid architecture can learn complex patterns on financial data which is hard to capture by stand-alone statistical models. The proposed framework is an important methodological contribution for economic issue forecasting systems, while initially it was designed for financial distress prediction. The fusion of feature extraction and learning of neural networks offers a good foundation for hybrid prediction models such as PLS-ANN.

Research Gap: The study was mainly limited to Altman Z-score variables and did not assess dimensionality reduction techniques i.e. PLS which can handle multi-collinearity involving economic factors.

2.2.4 Predicting Corporate Financial Distress Using a Hybrid Model of Artificial Immune System and Wavelet Neural Network

In 2024, Mehrabi et al. proposed a hybrid system to predict corporate financial distress of sample companies listed in Tehran Stock Exchange that is a combination of Artificial Immune Systems (AIS) and Wavelet Neural Networks (WNN).

Optimization and adaptation of features were done by using AIS mechanisms and for nonlinear pattern recognition and classification, WNN was used. Hybrid scheme was designed to take advantage of the complementary aspects of biological computation over the capabilities of a neural network learning to improve a prediction's accuracy.

Results of experiments showed it achieved better results than the standard machine learning methods. The Model has been successful in adapting, forecasting Wall Street and being robust in various financial scenarios.

The paper also corroborates the growing amount of research that integrates and combines the computational intelligence techniques successfully, thereby largely improving the prediction accuracy in complex financial scenarios. The results are applicable to economic forecasting, since in financial distress situations economic indicators are also found to have nonlinear and dynamic relationships.

Research Gap: Even though the model works well, it is hard to use in practice in large correspondent economic forecasting systems with complex calculations.

2.2.5 A Hybrid Network Analysis and Financial Distress Prediction Model

Taheri Kakhoda and Amiri (2024) came up with a hybrid financial distress prediction model, which is a combination of network analysis and machine learning classification algorithms. The study constructed the similarity network and correlation network based on the financial indicators, and added the features of the financial network to their prediction models.

This methodology was based on several detection methods, clustering and a number of classification techniques for financial distressed organisations. The hidden relationships among firms and financial variables were captured with network-based information.

In several prediction contexts, it was noticed that the use of features from the network, in particular, made a significant contribution to the classification results. The researchers found that the structure of the network has informative elements which have information about the model that may be ignored in the traditional feature sets.

This work is a step toward understanding the possible improvement of financial forecasting systems with the use of relational information. The results are in line with the trend of hybrid approaches towards model-based prediction.

Research Gap: Multi-collinearity between economic indicators is not addressed and the model is based on a network structure rather than using latent variable extraction.

2.3 Comparative Analysis of Reviewed Studies

Study	***Hybrid Technique***	***Dataset***	***Main Method***	***Key Findings***
Kumar et al. (2025)	*ANN + Ensemble Learning*	*Financial Distress Data*	*FINNET*	*High predictive performance*
Sruneethi & Vinodhkumar (2025)	*Network Analysis + ML*	*Financial Distress Dataset*	*RF, Voting Classifier*	*Improved prediction performance*
Nguyen et al. (2024)	*Altman Z-score + ANN*	*Vietnamese firms*	*MLP-ANN*	*98.4% accuracy*
Taheri Kakhoda & Amiri (2024)	*Network Analysis + ML*	*Financial Indicators*	*Hybrid Classification*	*Better predictive capability*
Mehrabi et al. (2024)	*AIS + WNN*	*Tehran Stock Exchange*	*AIS-WNN*	*Enhanced robustness and accuracy*

Table: Comparative Analysis of Reviewed Studies

2.4 Summary of Literature Review

The survey reveals that the traditional statistical approach has slowly shifted to a one or multiple machine learning models approach for financial and economic forecasting. Extensive studies corroborate that it is possible to achieve better predictive performance when using one or more of the following three approaches: feature extraction, optimization, network analysis or ensemble, or neural network architecture.

The dimensionality reduction technique, in particular, was being well supported in the works reviewed by Adisa et al. (2019) who gave the best evidence for the effectiveness of dimensionality reduction technique with ANN models. However, they do not take into account the relationships of target variables in their PCA-based approach. Similarly, others have combined one or more learning strategies, but generally used without a consideration of multicollinearity and the construction of latent variables.

Such restrictions give the opportunity of developing a 'Partial Least Squares–Artificial Neural Network (PLS-ANN)' Hybrid model. One of the key differences between PCA and PLS is that the components in PLS are designed to maximise the covariance (not correlation) between the predictor variables and the target variable and hence more useful components may be extracted for the prediction of the target. In order to increase the accuracy, efficiency and interpretability of the model, it is proposed to build a model combining the dimensionality reduction ability of PLS with the non-linear learning ability of ANN to make a more precise prediction of economic problems.

From this, it is concluded that there is validity in the use of hybrid PLS-ANN and from the literature, the theoretical basis for using the methodology presented in the following chapters could be established.

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CHAPTER 3 METHODOLOGY

3.1 Proposed Framework

In this study, a hybrid prediction system using two techniques, namely Partial Least Squares (PLS) regression and Artificial Neural Networks (ANN) is outlined for predicting economic problems from a mixture of financial and sales data. The framework's structure is designed to address the issues of high dimensional data sets, the imprecision of economic indicators, and non-linear relationships between the economic indicators and profitability.

The proposed framework is composed of 5 key steps: data pre-processing, feature learning, model training, model performance evaluation and comparison analysis. The financial data is first scrubbed and then transformed into a machine learning friendly data. Additional features are designed for other economic indicators that are more directly related to the business' economic performance. The data set is then pre-processed by standardization and transformation by Partial Least Squares (PLS) and latent variables are extracted, which will indicate the maximum covariance between predictor variables and target variables.

The extracted latent features under the condition are then given as the inputs to an Artificial Neural Network for "nonlinear learning and prediction". To validate the proposed model, the performance of the proposed model is compared with six benchmark models: ANN, Linear Regression, Ridge Regression, Lasso Regression, Elastic Net Regression and AdaBoost Regression. The aim of this framework is to illustrate the overall flow of the proposed framework.

3.2 Feature Learning Process

The proposed framework focuses on the important feature learning step that captures relevant information from the first financial variables. The ultimate goal is to get the original data to a compact representation that will contain information and will help to get a more accurate prediction and better generalization.

3.2.1 Feature Engineering

Some of the variables in the dataset are related to financial variables that are more readily available in the existing data sources, thus domain-specific variables can be created that will help to enhance the prediction power of the dataset. Any extra information that can be derived from these structured features regarding profitability, production efficiency and pricing strategies is an added value.

Revenue per Unit of the Lao PDR national economy was defined as:

$$Revenue_per_Unit = \frac{Sales}{Units\ Sold}$$

The cost per Unit is the same as:

$$Cost_per_Unit = \frac{COGS}{Units\ Sold}$$

The discount rate was computed as:

$$Discount_Rate = \frac{Discounts}{Gross\ Sales}$$

Profit Margin was determined by using the following method:

$$Margin = \frac{Profit}{Sales}$$

The relationships between various economic variables are highlighted in these engineered variables, which are not manifested explicitly in the original data set.

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3.2.2 Data Leakage Prevention

When information is directly related to the object variable, and is also one of the predictor variables. This leakage can lead to an unrealistic overestimation of performance and lack of generalization of the model in real-world applications.

Since the variables Sales, Cost of Goods Sold (COGS) and Gross Sales are directly mathematically related to Profit, the target variable in this study, the data from these variables were not included in the predictor set to avoid leakage by including data from all these variables in both the predictor set and the target variable.

This is to ensure meaningful patterns are being taught to the models and not information which they wouldn't observe during forecasting in a real context.

3.2.3 Feature Encoding

Some of the attributes are categorical, including Segment, Country, Product and Discount Band. To use the categorical values of variables as numbers for feeding machine learning algorithms, one-hot encoding has been used.

One Hot Encoding converts each category into a binary indicator variable that can be utilized by the models without any assumption of an ordering of the categories.

3.2.4 Feature Standardization

There are multiple measurement scales in the predictor variables. The numerical features were all standardized by using the Standard Scaler, removing the scale information bias and for efficient optimization.

One type of standardization of features is a transformation of each feature according to:

$$z = \frac{x - \mu}{\sigma}$$

where:

- x represents the original value
- μ denotes the mean
- σ denotes the standard deviation.

Standardization makes features uniform so that they all can play an equal role in training the model.

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3.2.5 Partial Least Squares Feature Extraction

Here partial least squares feature extraction technique is used. The partial least squares extraction method is as follows: To reduce the dimensionality without loss of information pertinent to the target variable, a dimensionality reduction technique was used, namely, Partial Least Squares (PLS). While Principal Component Analysis (PCA) is based on the variance in predictors, PLS is based on covariance between the predictors and the response variable in order to find latent variables that maximize covariance in the model.

PLS factors the predictor matrix (X) and response vector (Y) into latent structures:

$$X = TP^T + E$$

$$Y = UQ^T + F$$

where:
The matrices (T) and (U) are matrices of scores,
The matrix (P) is a loading matrix, and the matrix (Q) is a loading matrix.
The resulting matrices are shown: (E) and (F).

Various PLS-models with 2-25 latent variables were evaluated to achieve the optimum model, which was obtained when the lowest root mean square difference (rms) was attained: 2 latent variables were tested in model-1 and then gradually one, two, quarter and in sixteen steps, latent variables were added up to the model which resulted from the 25th step. The optimum arrangements of the components that maximized the coefficient of determination (R2) were chosen. After the experimental evaluation, 14 latent components were considered for further modelling. Following experimental evaluation, 24 latent components were found to be optimal and used for modeling.

3.3 Model Training Process

3.3.1 Train-Test Splitting

The data was divided into training and testing data where 80% of the data is used as training data and 20% as testing data. The model was learned with the training set and the performance of the predictive model was tested with the testing set. To ensure consistency of experimental results, fixed random state was used for splitting process.

3.3.2 Hyperparameter Configuration

The hyperparameters were carefully chosen for each model. The proposed PLS-ANN model is based on the following equation.
Used a deep neural network with multiple hidden layers that used the LeakyReLU activation function. Dropout regularization along with activation functions were used. In the case of regularized regression models, suitable penalty parameters were set to balance the model. The complexity and the accuracy of the prediction. AdaBoost Regression was based on multiple weak learners to create a strong learner ensemble prediction model.

3.3.3 Optimization Procedure

The Adam optimization algorithm was used for optimization of the Artificial Neural Network models. Adam, a combination of the advantage of momentum based optimization with adaptive learning rates and can learn efficient convergence.
The value of the loss function while training was Mean Squared Error (MSE):

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

The training was carried out in several epochs using mini batches to assure stable parameter updates and enhanced generalization.

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3.4 Performance Evaluation and Comparative Analysis

3.4.1 Mean Absolute Error (MAE)

Mean Absolute Error measures the average absolute difference between actual and predicted values:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

Lower MAE values indicate higher prediction accuracy.

3.4.2 Coefficient of Determination (R^2)

The coefficient of determination evaluates the proportion of variance explained by the model:

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}$$

Higher R^2 values indicate stronger predictive capability.

3.5 Implementation Details

3.5.1 Data Preprocessing and Feature Engineering

The implementation starts with loading financial data from an excel file. Machine readable formats are used to represent the data for numerical variables, and missing values are not included. Next, various other financial indicators are created such as Revenue per Unit, Cost per Unit and Discount Rate, and Profit Margin, through feature engineering. Once feature engineering is done, variables that result in data leakage are dropped. A matrix of dimension 2 x 2 is used to code categorical variables. All the numerical features are scaled using Standard Scaler, while using One-Hot Encoding for the Categorical feature. The data is then latentized by means of Partial Least Squares for training the model.

3.5.2 Training Configuration

The experiments were conducted in python and some machine learning libraries including Scikit learn, TensorFlow, Keras, NumPy, Pandas and Matplotlib. The data set was divided into train (80%) and test (20%) sets in random manner. The predictor variables were standardized by using Standard Scaler. Optimization of components was carried out through evaluating the PLS components. The sizes can vary between 2 and 25, and the one with the highest R^2 is selected. For ANN type models training was performed using the Adam optimizer, learning rate of 0.01. Networks in which the model was trained for 300 epochs with a batch size of 32. The drop-out layers were added to minimize avoid overfitting and enhance generalization.

3.6 Models Used

For detailed comparison and analysis of the suggested hybrid model's effectiveness, six machine learning and the same training and testing data were used with the statistical regression algorithms and compared datasets. The main prediction model for this research was Artificial Neural Network (ANN), Linear Regression, Ridge Regression, etc. and the suggested model of this research was Partial Least Squares–Artificial Neural Network (PLS-ANN) model. Lasso Regression, Elastic Net Regression and AdaBoost Regression were used as benchmark models. To ensure fairness and unbiased results comparison, all models were trained using the same data pre-processing procedure.

3.6.1 Proposed Hybrid PLS–ANN Model

Architecture

The proposed hybrid model is a dimensionality reduction capability of Partial Least Squares (PLS) combined with the nonlinear learning ability of an Artificial Neural Network (ANN) on which the regression was performed. The main reason for the above-mentioned integration of the two methods is to overcome the multicollinearity which exists between the variables of financial indicators. At the same time, it is able to capture complex nonlinear relationships that exist within economic datasets. At first, predictor matrix 'X' is standardized and then a lower dimensional latent space is created by applying principal components analysis (PCA).

Partial Least Squares Regression (PLSR). PLS is different from Principal Component Analysis (PCA) since it does not maximize the variability of the predictor variables, but rather the covariance between predictor variables and the response variable.

The decomposition process can be represented as

$$X = TP^T + E$$

$$Y = UQ^T + F$$

where

• T denotes the latent score matrix,

• P denotes predictor loadings,

• U denotes response scores,

• Q denotes response loadings,

• E and represent residual matrices

The derived latent variables give a condensed version of the financial data without much loss of information that is relevant to prediction. The transformed feature matrix is then fed as an input in an Artificial Neural Network.

Neural Network Structure

The ANN consists of a fully connected feed-forward architecture composed of:

- Input Layer: 24 PLS latent variables
- Hidden Layer 1: 128 neurons
- Hidden Layer 2: 64 neurons
- Hidden Layer 3: 32 neurons
- Hidden Layer 4: 16 neurons
- Hidden Layer 5: 16 neurons
- Output Layer: 1 neuron

LeakyReLU activation functions are employed within hidden layers to prevent dead neurons and improve gradient propagation.

The activation function is defined as

$$f(x) = \{x, x>00.01x, x \leq 0 \}$$

Dropout regularization is incorporated after the first two hidden layers to reduce overfitting and

improve model generalization.

Training Procedure

The network parameters are optimized using the Adam optimization algorithm with Mean Squared Error (MSE) as the objective function:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

where and are the true and forecast economic value, respectively.

The PLS-ANN model enjoys at the same time the advantages of latent feature extraction and nonlinear approximation, which makes it very appropriate to economic prediction problems with correlated financial indicators.

3.6.2 Artificial Neural Network (ANN)

Architecture

The major baseline Deep Learning model is Artificial Neural Network. The ANN is fed with standardized predictor variables without reducing the dimensionality of the variables.

The network is composed of several closely connected layers, and learns a nonlinear relationship between the economic indicators and the target variable by means of iterative optimization.

Mathematical Representation

Each neuron computes

$$z = \sum_{i=1}^{n} w_i x_i + b$$

where

- Wi represents connection weights,
- Xi denotes input features,
- *b* represents the bias term.

LeakyReLU activation function is used to generate the output of the neuron.

$$a = f(z)$$

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A linear activation function is used to predict the economic value in the final output layer.

Training Process

he Adam optimizer with Mean Squared Error loss is used to train the ANN. Back Propagation is an iterative method, which updates network parameters until convergence. While ANN is good in modelling nonlinear relationship, all the original variables may contain redundant information and multicollinearity which affects the prediction accuracy of the model compared to the proposed hybrid model.

3.6.3 Linear Regression

Architecture

Linear Regression is one of the most popular statistical prediction methods and is used as a reference model. The model assumes that there is a linear relationship between the predictor variables and the target variable.

The regression equation is expressed as

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon$$

where

- β0 denotes the intercept,
- βi represents regression coefficients,
- Xi denotes predictor variables,
- ε represents random error.

Training Procedure

The parameters of the model are estimated by the Ordinary Least Squares (OLS) method that minimizes the Residual Sum of Squares.

$$RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

Linear Regression has high interpretability but there is an assumption that the variables are linear, therefore it is not suitable for complex economic prediction task.

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3.6.4 Ridge Regression

Architecture

Ridge Regression is an extension of Linear Regression with the addition of L2 regularization which shrinks the magnitude of the coefficients and thus mitigates multicollinearity.

The optimization objective is

$$L = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

where

λ

controls the strength of regularization.

Training Procedure

The optimization minimizes both prediction error and coefficient size, to obtain better stability of the model when the predictor variables are highly correlated.

Ridge Regression is good at dealing with overfitting and does not discard any of the predictor variables.

3.6.5 Lasso Regression

Architecture

Lasso Regression adds L1 regularization to regression, and allows automatic feature selection via coefficient shrinkage.

Its objective function is

$$L = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

The L1 penalty pushes less important coefficients towards zero, which makes the prediction model simpler.

Training Procedure

Iterative optimization is done to get convergence. Variables that do not significantly contribute to the prediction equation are excluded from the equation. Lasso Regression offers better interpretability and

also makes the model model simpler.

3.6.6 Elastic Net Regression

Architecture

Elastic Net Regression is a hybrid of Ridge and Lasso Regression, offering the benefits of both L1 and L2 penalty.

It aims to optimise the following objective:

$$L = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda_1 \sum |\beta_j| + \lambda_2 \sum \beta_j^2$$

The use of penalties allows for both feature selection and coefficient shrinkage.

Training Procedure

Elastic Net is best used for data sets with high correlation between variables as compared to Ridge or Lasso, it gives a better balance between sparsity and regularization.

Coefficient values are found during the optimization process to minimize the prediction error while controlling the model complexity.

3.6.7 AdaBoost Regression

Architecture

AdaBoost Regression is an Ensemble Method that is used to build a predictive model by combining multiple weak regression learners. AdaBoost is a method that trains a series of weak learners, each of which aims to correct the errors made by the previous learners.

The last prediction is calculated as

$$F(x) = \sum_{m=1}^{M} \alpha_m h_m(x)$$

where

- hm (x) denotes the mth weak learner,
- αm represents the weight assigned to each learner,
- M denotes the total number of boosting iterations.

Training Procedure

The algorithm starts with equal sample weights and, based on the prediction errors, gradually adjusts the weights. Difficult to predict samples are weighted more in later iterations, which enables the ensemble to concentrate on difficult observations.

AdaBoost is a combination of several weak regressors and can be used to improve the prediction accuracy and decrease the bias; hence, it is used as a reference for the proposed hybrid PLS-ANN framework.

CHAPTER 4 - RESULTS AND DISCUSSION

4.1 Introduction

This chapter provides a detailed explanation of the empirical findings generated by the suggested hybrid forecasting model based on Partial Least Squares (PLS) regression and Artificial Neural Networks (ANN). The primary goal of this chapter is to assess the usefulness of hybrid modeling method in predicting the economic variable (profit) from a structured set of financial and operational data.

The results are reported in a sequential analytical structure, starting with exploratory data analysis (EDA), then preprocessing outcomes, extraction of latent features using PLS, training behavior of ANNs and finally the comparative performance evaluation with multiple benchmark machine learning models. The results are also discussed from the perspective of model behavior, the characteristics of the economic data, and the efficiency of predictive learning.

4.2 Results and Performance Evaluation

4.2.1 Overall Model Performance

For the purpose of checking the performance of the proposed prediction framework, 7 regression models were developed and compared using the same training and testing data set. The coefficient of determination (R2), Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) were three widely used regression evaluation methods that were used to assess the performance of the regression. The higher the R2, the better, as are RMSE and MAE values.

The data set has been pre-processed extensively before training the models including removal of missing values, feature engineering, data leakage prevention, one hot encoding, feature standardization and partial least squares (PLS) feature extraction. The models were then created based on the modified features.

The results of all the models implemented are summarized in table X.

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Model	*R^2 Score*	*RMSE*	*MAE*
PLS + Artificial Neural Network (Proposed Model)	*0.9757*	*1084.98*	*769.55*
AdaBoost Regression	*0.9233*	*1928.72*	*1628.60*
Linear Regression	*0.8843*	*2369.17*	*1871.82*
Lasso Regression	*0.8791*	*2421.54*	*1892.75*
Ridge Regression	*0.8531*	*2669.39*	*2034.07*
Elastic Net Regression	*0.8414*	*2773.54*	*2126.74*
Artificial Neural Network (Without PLS)	*0.7652*	*3374.48*	*2564.97*

Table: Overall Model Performance

Based on the results of the experiments, it was concluded that the proposed PLS-ANN hybrid model was superior to all the baseline machine learning models. When the R2 value of the model was 0.9757, the proposed model explained about 97.57% of the variance in the values of profit, and the model had the least prediction error among all the competing models.

The results of the suggested PLS-ANN model are shown in this section. In this section, the performance of the proposed PLS-ANN model is presented.

Graphs:

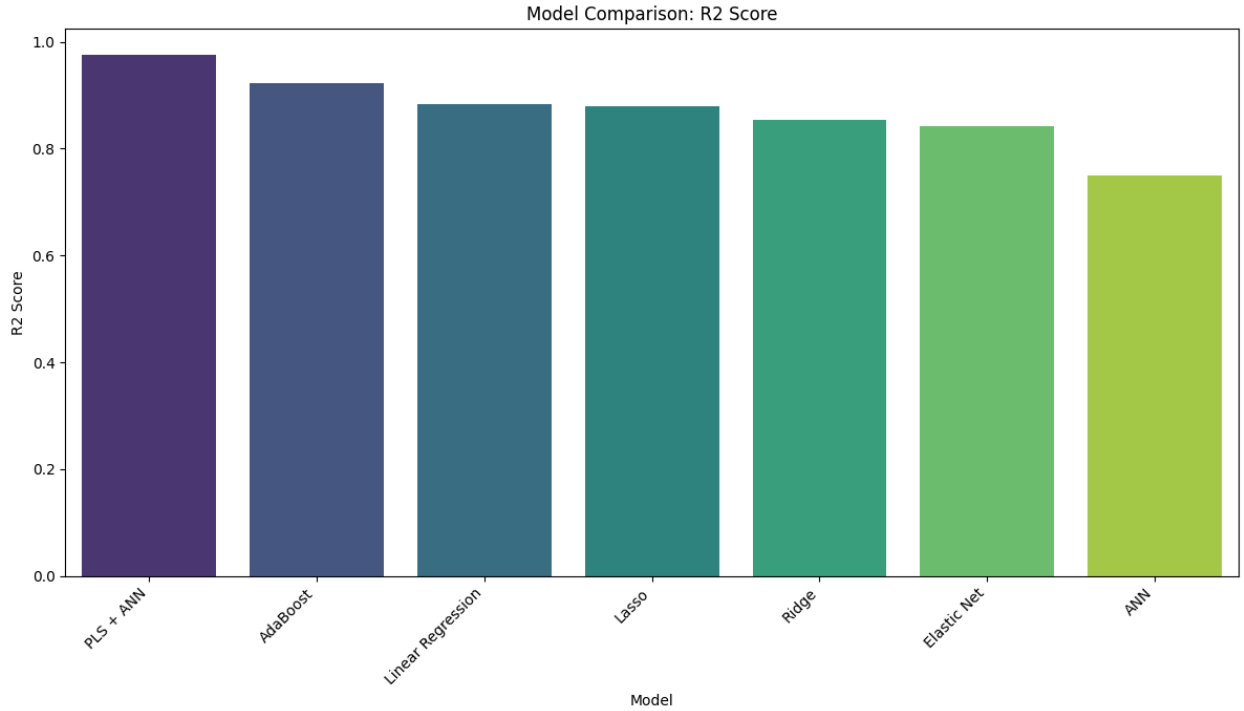


Fig: 4.2.1.1 Model Comparison R^2 Score

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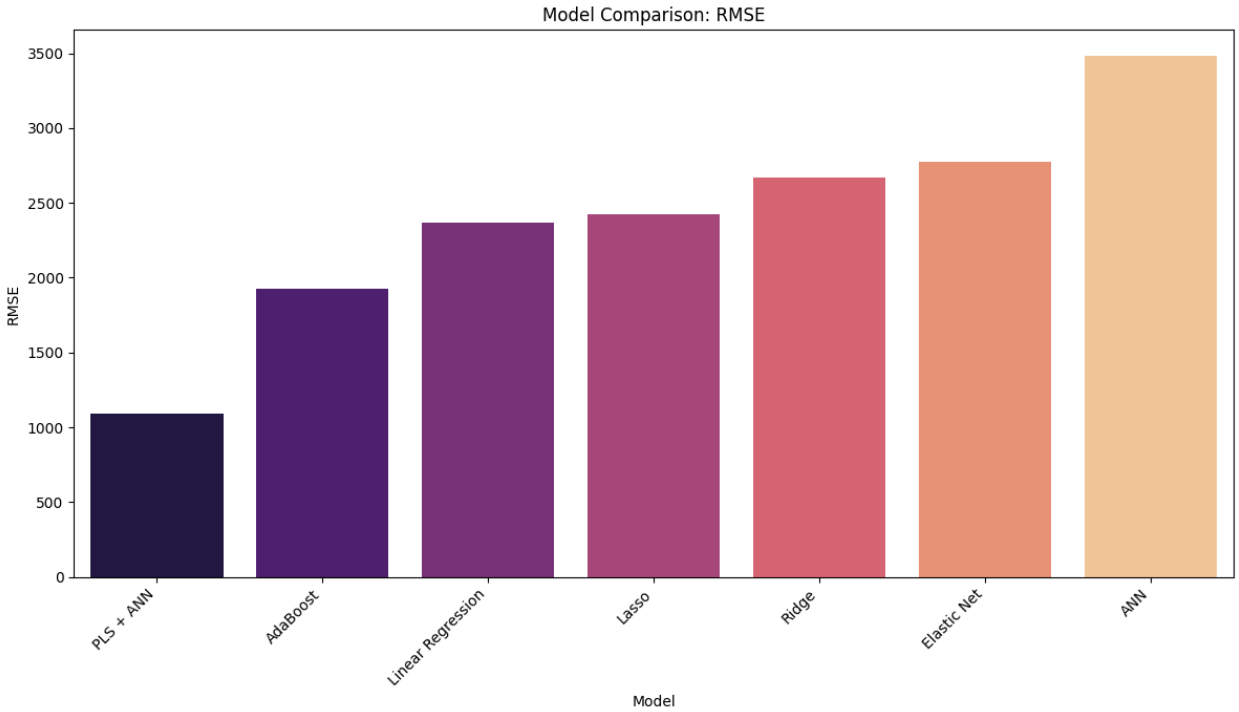


Fig: 4.2.1.2 Model Comparison RMSE

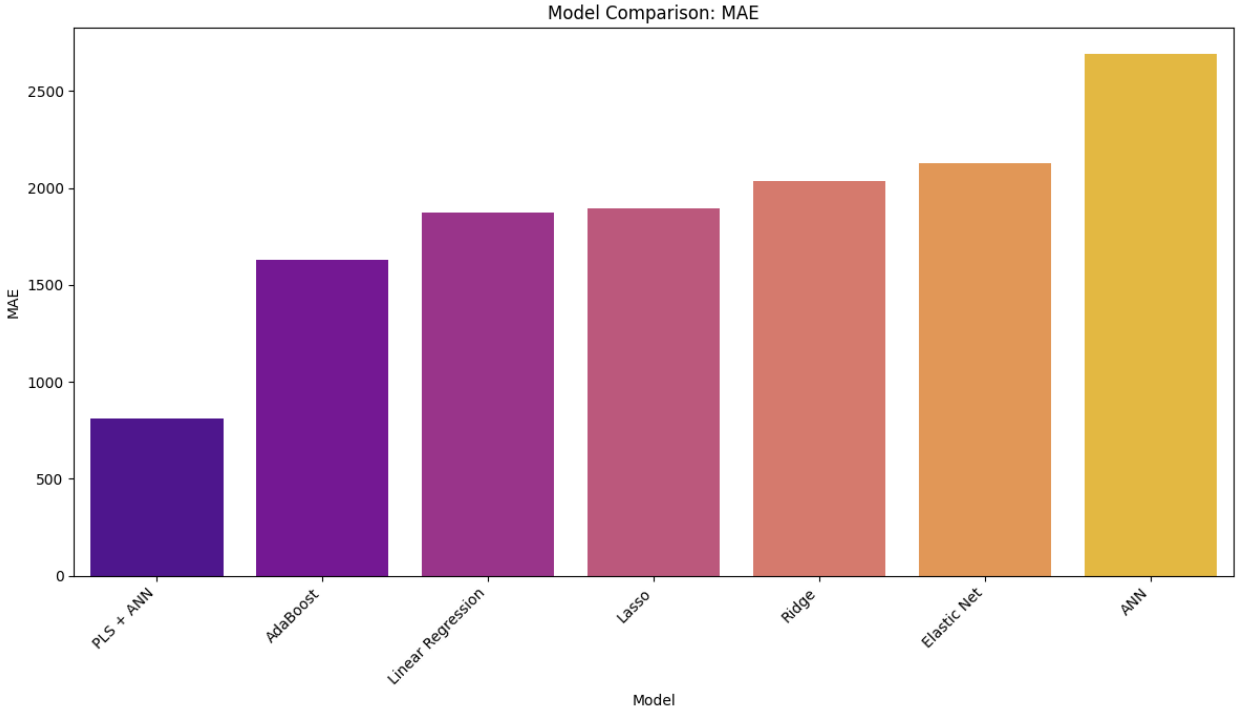


Fig: 4.2.1.3 Model Comparison MAE

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4.2.2 PLS + Artificial Neural Network

Deep Artificial Neural Network (ANN) structure for feature extraction process. Processing times involved two phases: a normalization operation was first performed on all the numerical variables and a one-hot encoding procedure was secondly performed on categorical variables. Next, PLS derived latent components, which are able to maximise the covariance between the predictors and the target variable, so as to minimise the effect of "overlapping information" and "multi-collinearity" prior to the training of the neural network.

The ANN architecture consisted of multiple hidden layers of 128 neurons, 64 neurons, 32 neurons, 16 neurons and 16 neurons, with LeakyReLU activation functions, and the dropout regularization method was used to prevent overfitting. Used validation loss to optimize the model with Adam optimizer and early stopping.

The proposed model was able to achieve the following results:

$R^2 = 0.9757$
RMSE = 1084.98
MAE = 769.55

The scatter plot of Actual versus Predicted depicts a good consistency of prediction, with the predicted values being very close to the ideal line of 45°. Also, the residuals are approximately centered, indicating that there is little systematic prediction bias and good generalization ability.

The superior performance indicates that PLS can extract highly informative latent variables, and it is very useful for the learning ability of neural network.

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Graphs:

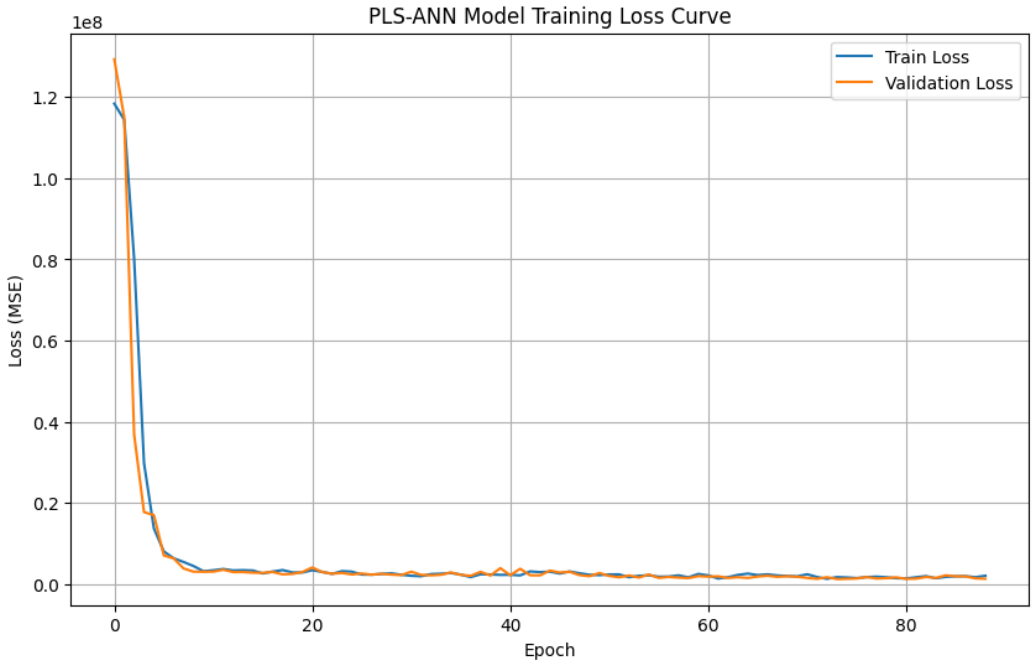


Fig: 4.2.2.1 PLS-ANN Model Training Loss Curve

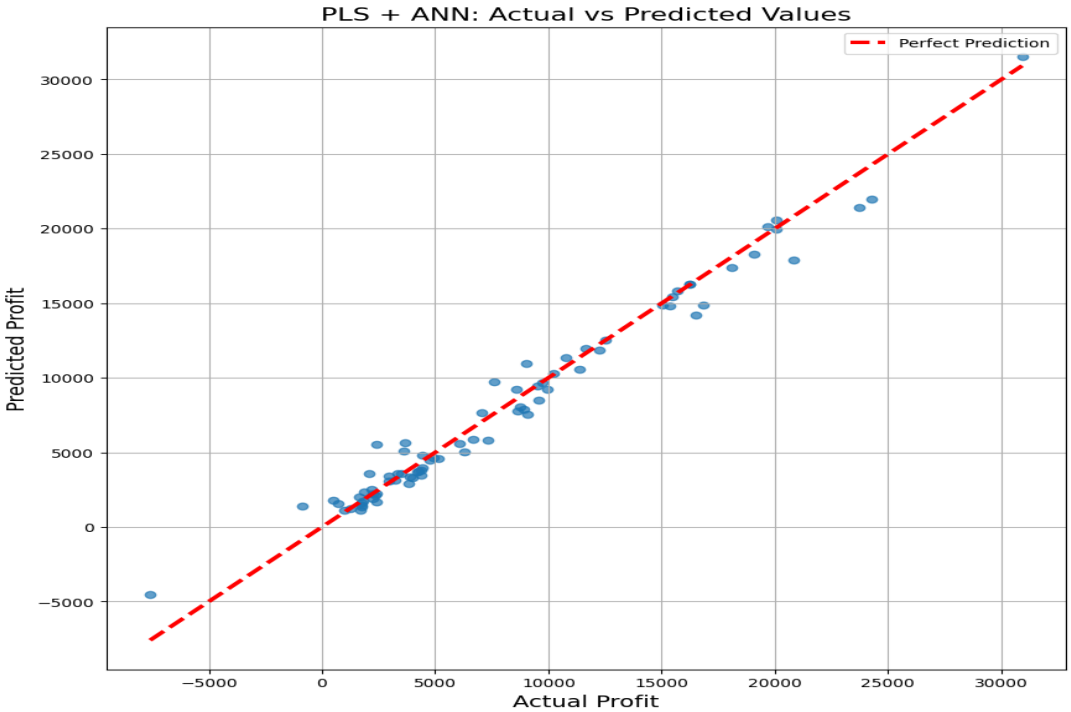


Fig: 4.2.2.2 PLS + ANN: Actual vs Predicted Values

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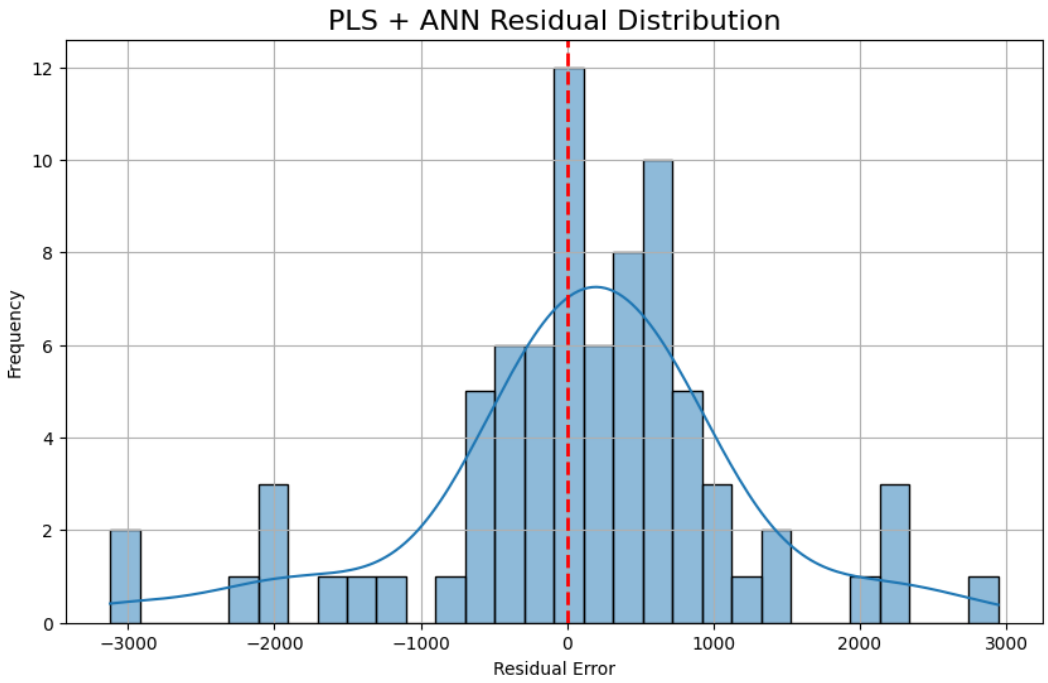


Fig: 4.2.2.3 PLS + ANN Residual Distribution

4.2.3 Linear Regression

Linear regression is among the easiest types of regression.

Basic model: Linear Regression.

The model achieved:

$R^2 = 0.8843$

RMSE = 2369.17

MAE = 1871.82

Linear Regression showed the overall trend of the explanatory variables to profit, but the prediction power of this approach was lower than the proposed PLS-ANN approach. The relative magnitude of the prediction errors indicate that there are nonlinearities between financial variables and profit which cannot be explained by a linear function.

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Graphs:

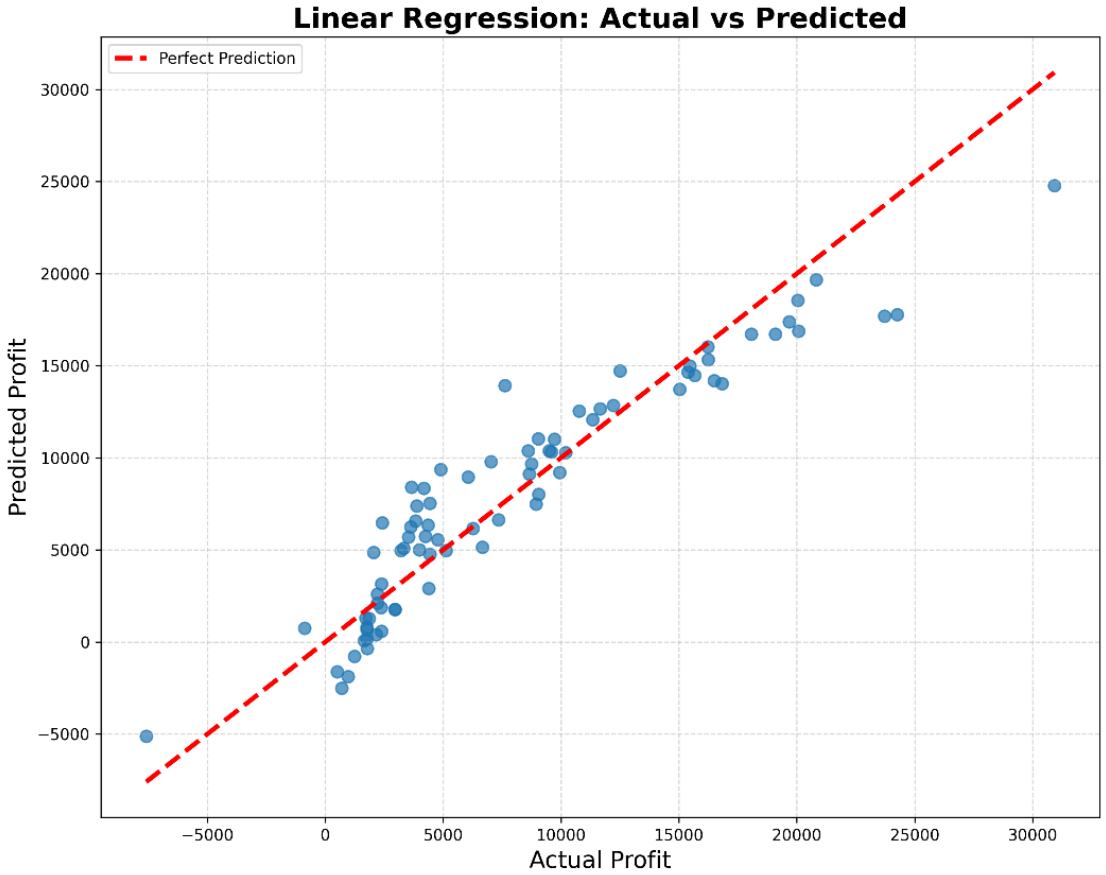


Fig: 4.2.3.1 Linear Regression: Actual vs Predicted

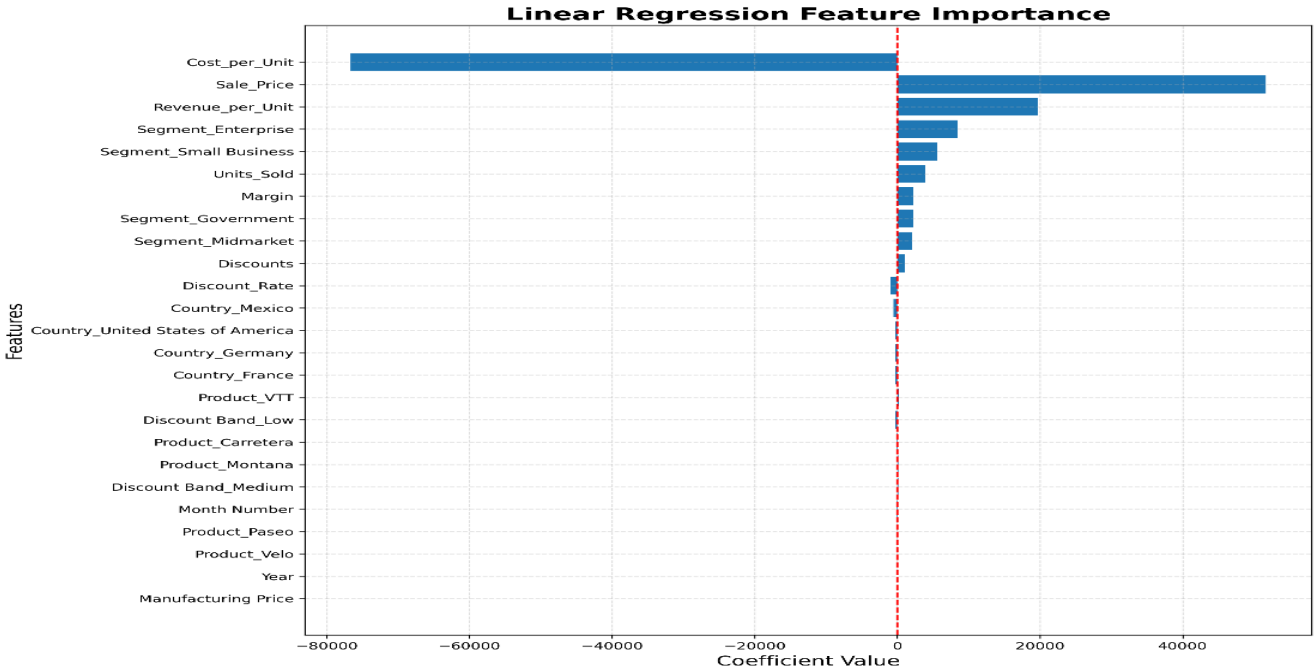


Fig: 4.2.3.2 Linear Regression Feature Importance

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4.2.4 Performance of AdaBoost Regression.

Let's look at 4.X.4 Performance of AdaBoost Regression.

The second-best overall performance was obtained by the AdaBoost Regression method.

It was judged by the following:

$R^2 = 0.9233$

RMSE = 1928.72

MAE = 1628.60

The feature importance analysis also demonstrated that the values of the most important financial indicators that influence the prediction of profit were successfully identified by AdaBoost. Although AdaBoost has a good ensemble learning potential, it still produced relatively higher prediction errors compared with the proposed hybrid model, which has demonstrated that the use of nonlinear latent representations learnt by PLS provides additional prediction information to the ensemble boosting than ensemble boosting alone.

Graphs:

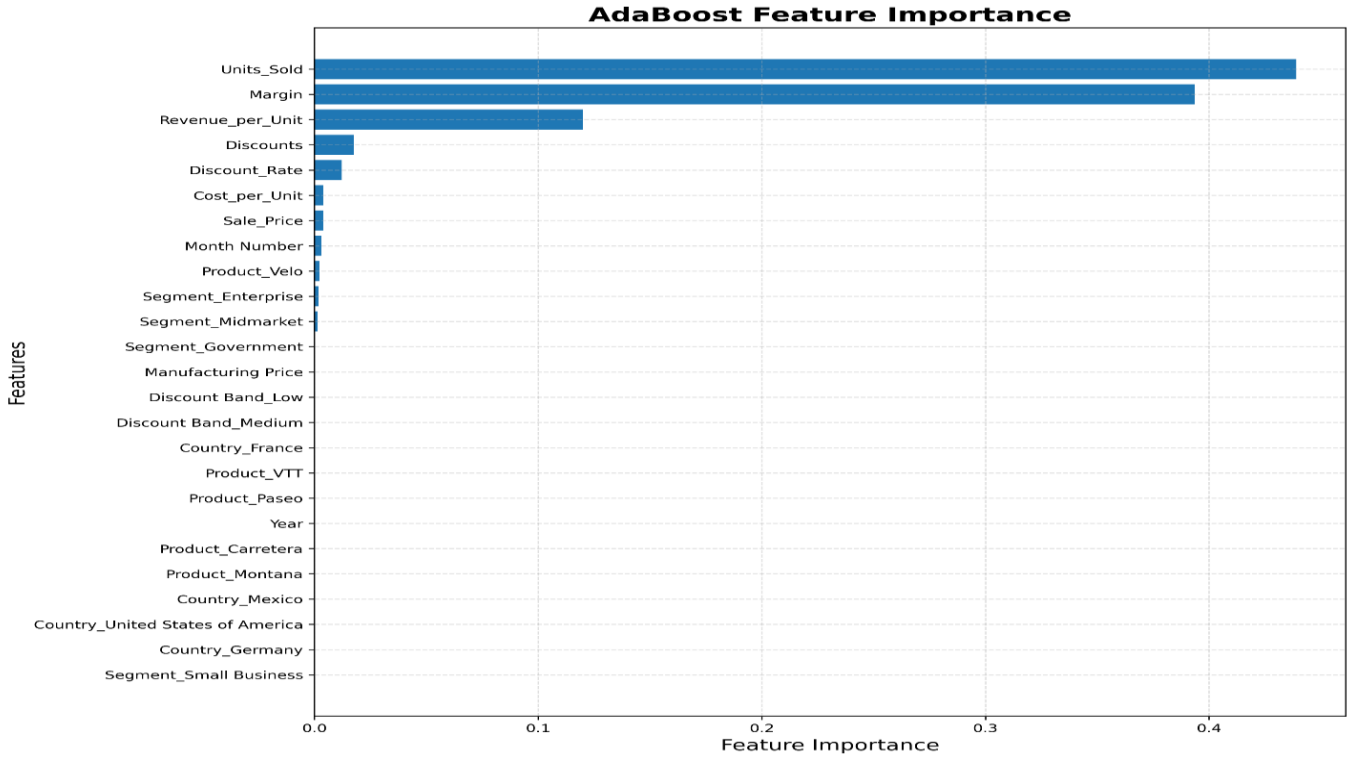


Fig: 4.2.4.1 AdaBoost Feature Importance

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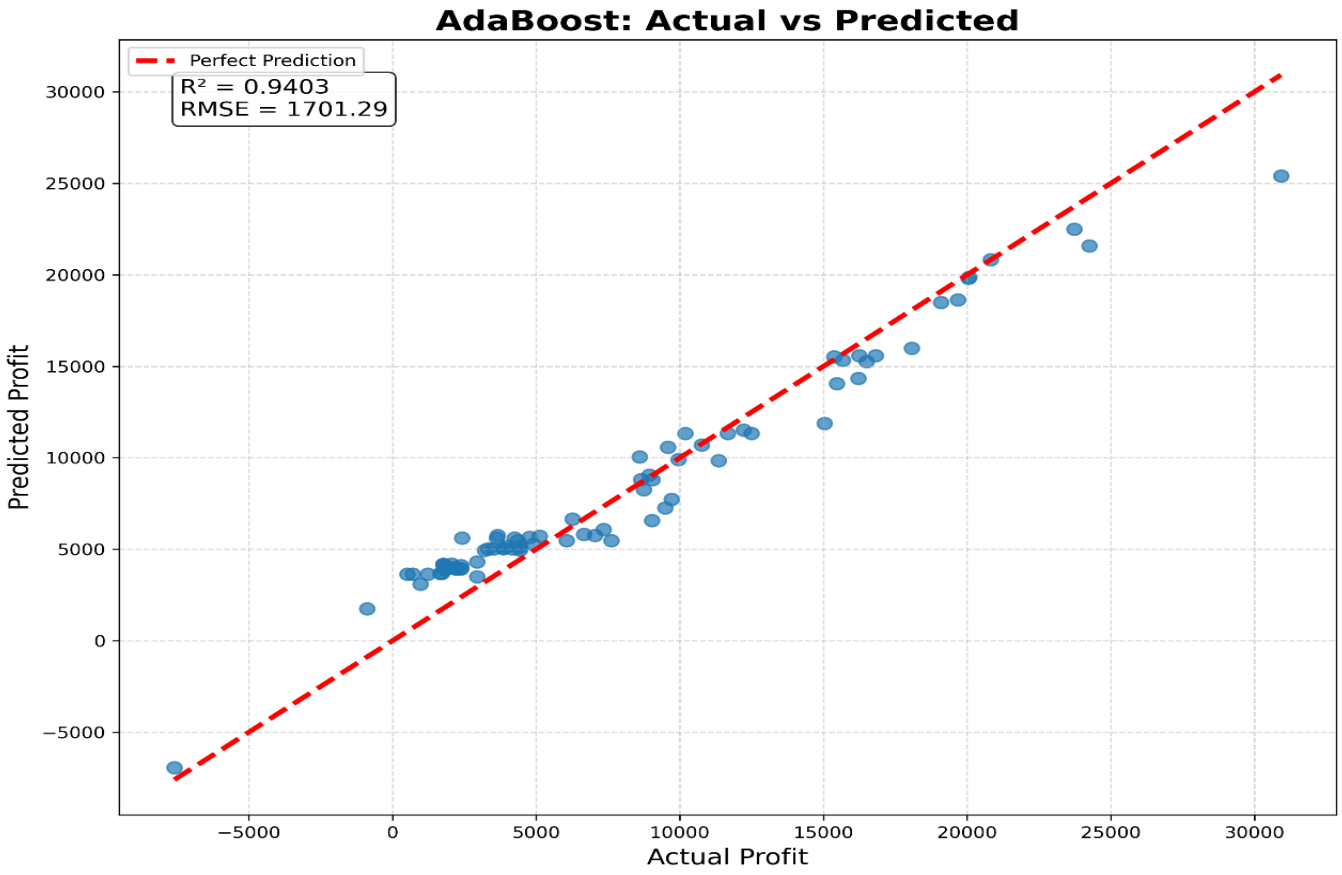


Fig: 4.2.4.2 AdaBoost: Actual vs Predicted

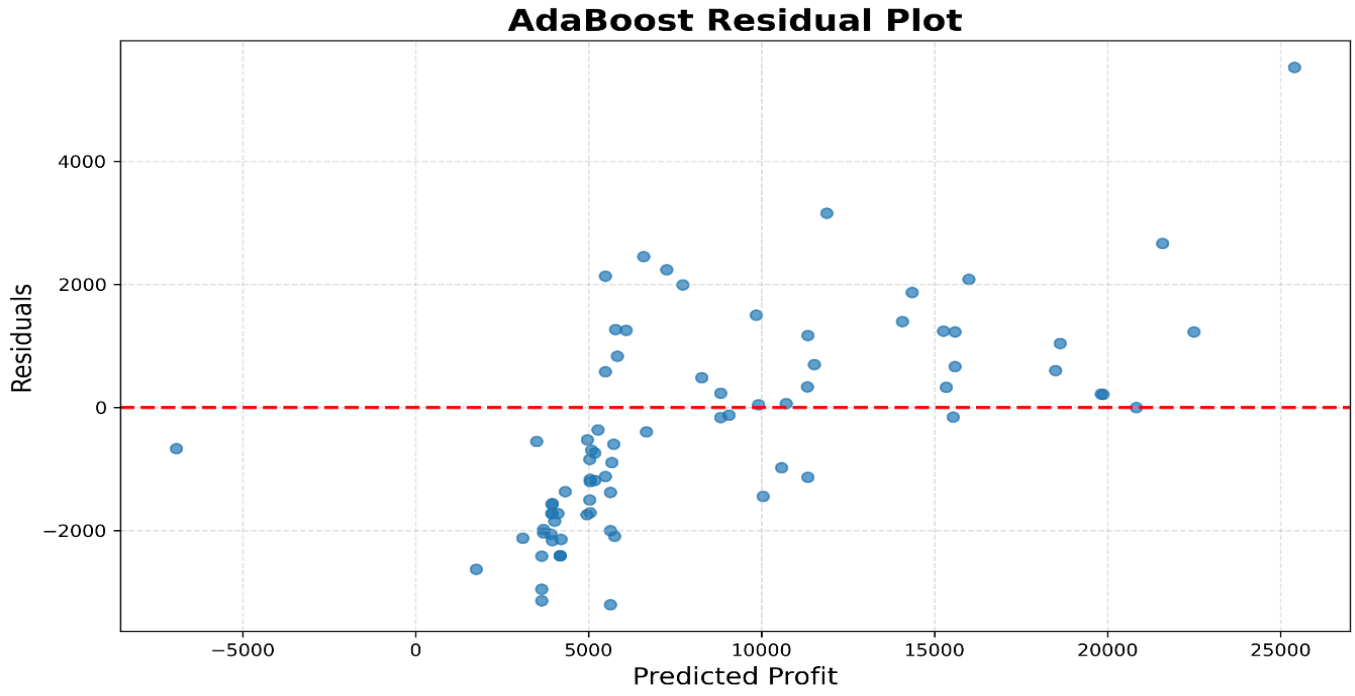


Fig: 4.2.4.3 AdaBoost Residual Plot

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4.2.5 Elastic Net Regression:

Elastic Net is a mix of L1 and L2 regularization to get a combination of feature selection and shrinkage of coefficients.

The model obtained:

$R^2 = 0.8414$
RMSE = 2773.54
MAE = 2126.74

The coefficient shrinkage plot indicates that the coefficients are shrinking more and more with increasing regularization strength. Elastic Net can also be very useful to alleviate the effects of multicollinearity, but unlike nonlinear methods, it is linear and so may not be suitable for modelling more complex financial relationships (where the prediction accuracy is greater).

Graphs:

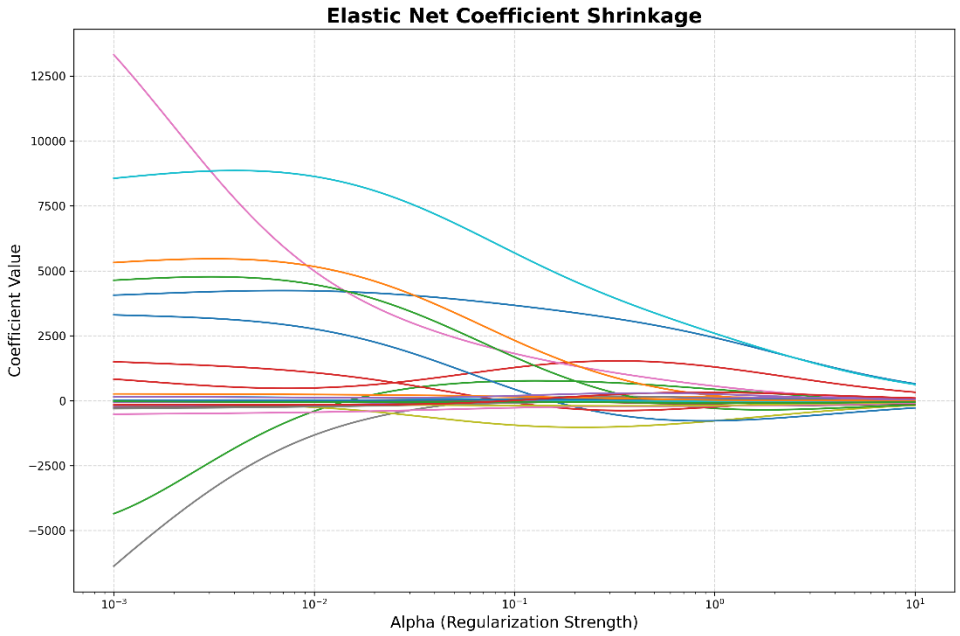


Fig: 4.2.5.1 Elastic Net Coefficient Shrinkage

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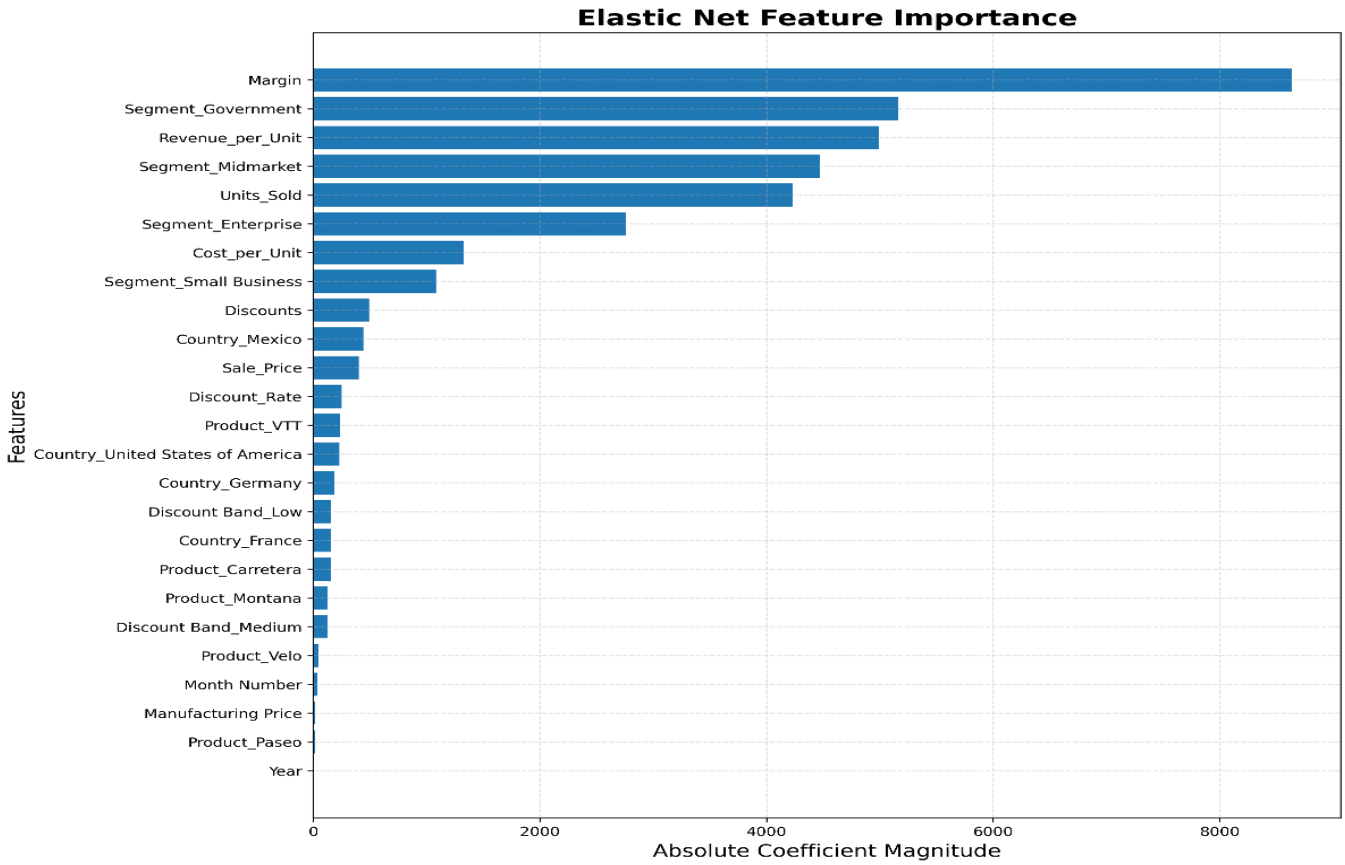


Fig 4.2.5.2 Elastic Net Feature Importance

4.2.6 Performance of Lasso Regression

To encourage sparsity (simple feature selection) and reduce overfitting, Lasso Regression introduces L1 regularization.

The results that were obtained were:

$R^2 = 0.8791$

RMSE = 2421.54

MAE = 1892.75

As the regularization value increased, some of the features' coefficients shrank to 0, indicating successful variable selection in the coefficient shrinkage analysis. Some variables were dropped from the analysis as they did not provide much information but maybe slightly decreased the predictive capabilities of the model from Linear Regression.

Graphs:

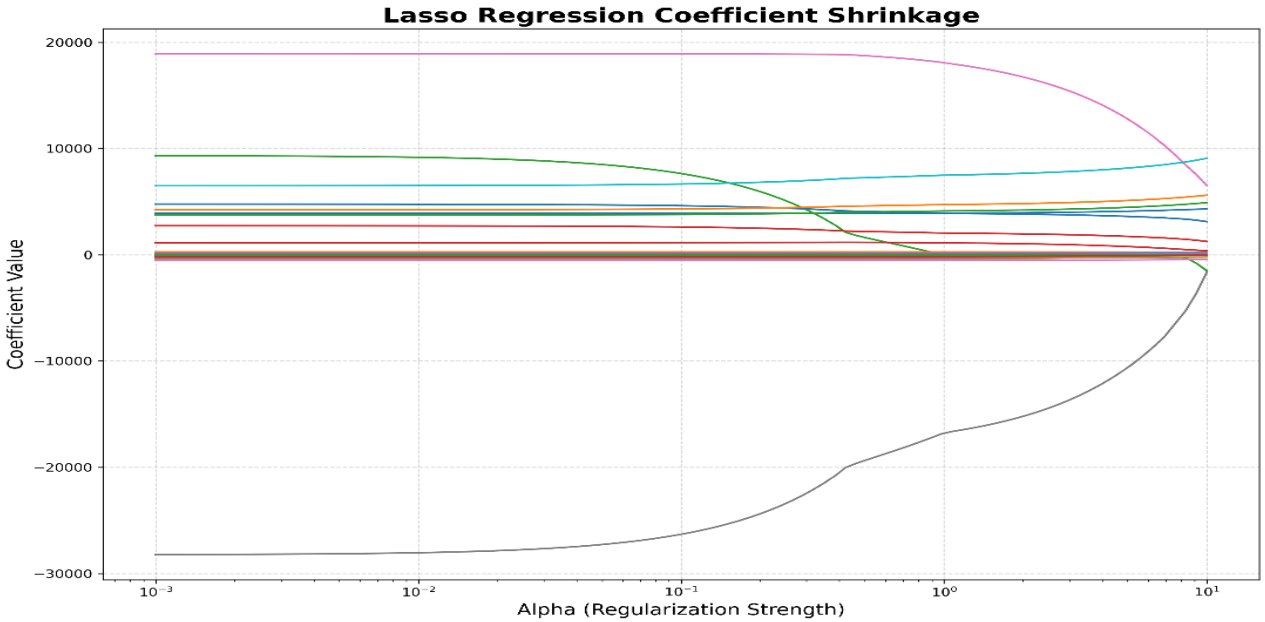


Fig 4.2.6.1 Lasso Regression Coefficient Shrinkage

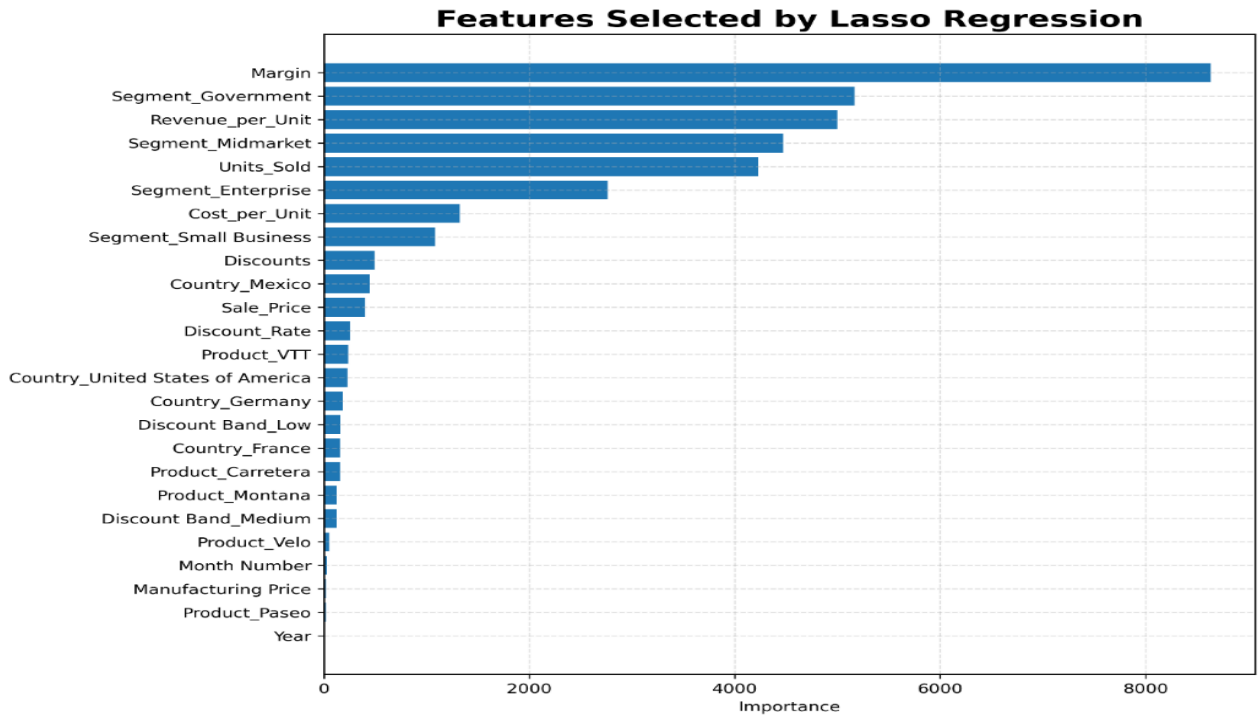


Fig: 4.2.6.2 Features Selected by Lasso Regression

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4.2.7 Performance of Ridge Regression.

Compute the performance of Ridge Regression. Ridge Regression is method that stabilises the coefficients of the model if multicollinearity occurs, using L2 regularisation.

Its performance was:

$R^2 = 0.8531$

RMSE = 2669.39

MAE = 2034.07

The coefficient shrinkage plot yielded a gradual decrease in the magnitude of the coefficients but no coefficients being set to zero. Ridge was an improvement in terms of stability but, since it is a fundamental linear model, it was not as accurate as ensemble and deep learning models.

Graphs:

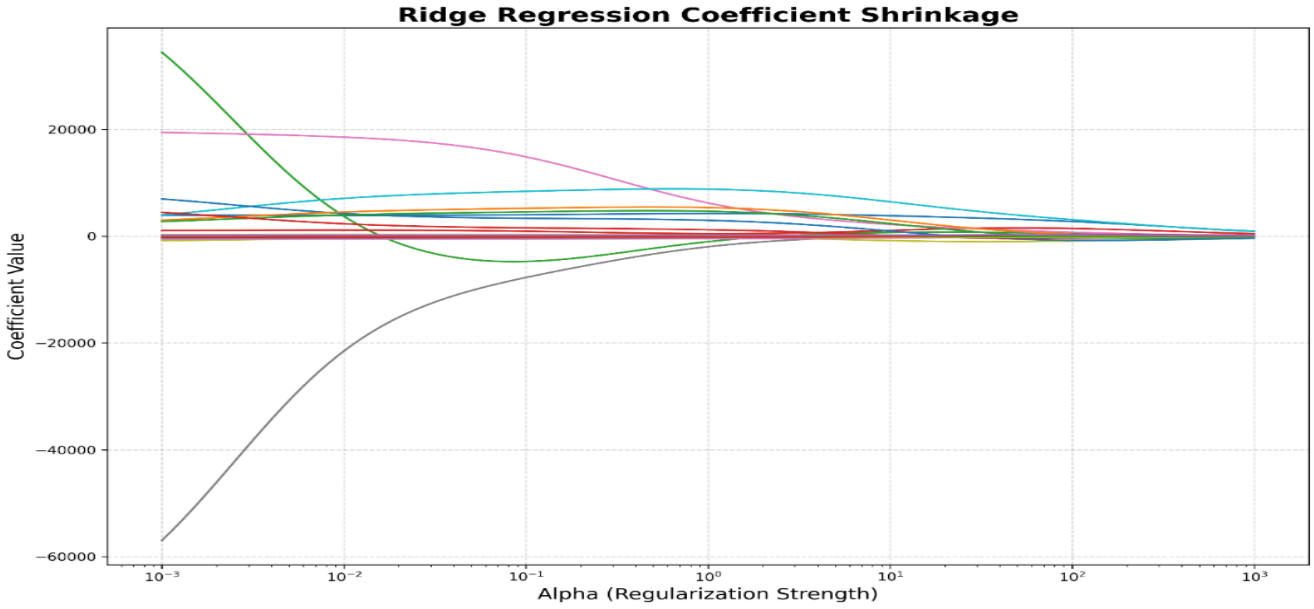


Fig: 4.2.7.1 Ridge Regression Coefficient Shrinkage

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4.2.8 Performance of Standalone Artificial Neural Network

A stand-alone Artificial Neural Network, where no latent feature learning was performed, was also evaluated to gain insight into the impact of latent feature learning.

The model achieved:
$R^2 = 0.7652$
RMSE = 3374.48
MAE = 2564.97

Standalone ANN was the less accurate of all the approaches implemented although a deep neural architecture was used. The results indicate that taking standardized high dimensional features directly does not yield a good effect as taking low dimensional latent features obtained by Partial Least Squares.

The substantial improvement when PLS and ANN are fused together demonstrates the fact that dimensionality reduction and latent feature extraction make a great improvement in the efficiency of learning and prediction of neural networks.

4.2.9 Comparative Discussion

The comparative analysis was clearly indicated that the proposed model (PLS-ANN) is superior to all the six benchmark model in all the evaluation metrics.

The proposed framework outperformed the best base model (AdaBoost) by increasing the R2 value by 0.9757 against 0.9233, while decreasing the RMSE by 1084.98 against 1928.72 and the MAE by 769.55 against 1628.60.

CHAPTER 5- CONCLUSION

5.1 Introduction

Economic forecasting was made more difficult by the nonlinear and complex nature of financial data in today's era that needed to be analyzed to make predictions. Conventional statistical methods can be ineffective in capturing hidden interactions among economic time series while individual machine learning can be ineffective because of the redundant information or multicollinearity. This research proposed the hybrid prediction system to deal with these limitations using the feature extraction technique named Partial Least Squares (PLS) and Artificial Neural Networks (ANN) for economic issue prediction.

The proposed framework combines the dimensionality reduction capacity of PLS and non-linear approximation capacity of the deep neural network, and it can learn dimension reduced latent representation preserving the most relevant information for the prediction target. Extensive experiments were carried out to evaluate the effectiveness of the proposed approach and the performance of the proposed approach was compared with several popular regression algorithms.

5.2 Summary of Findings

The experimental investigation shows that PLS-ANN combination can significantly enhance the predictive performance compared to single machine learning and statistical regression methods.
The proposed hybrid model resulted in:

- R^2 Score = 0.9757
- RMSE = 1084.98
- MAE = 769.55

The results show that the variance explained by the proposed framework is around 97.57% and the lowest prediction error among all the models evaluated.

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Comparative analysis further revealed that:

Model	***R²***
PLS-ANN (Proposed)	***0.9757***
AdaBoost	*0.9233*
Linear Regression	*0.8843*
Lasso Regression	*0.8791*
Ridge Regression	*0.8531*
Elastic Net	*0.8414*
ANN without PLS	*0.7652*

Table: Comparative analysis

The results show that the use of latent feature extraction is most beneficial in the learning process of the neural networks.

Especially the comparison with the standalone ANN is very interesting. Though it might be possible that ANN can be trained to learn the nonlinear relationships, direct training of high dimensional financial variables did not get good performance. Multicollinearity and redundant information were greatly reduced following PLS transformation and the neural network was focused on very informative LVs. This led to a considerable improvement in predictive accuracy and model stability.

5.3 Significance of the Study

This study is beneficial from both academic and practical aspects for the field of intelligent economic forecasting.

The methodological aspect of the study is that by using supervised latent feature extraction in conjunction with deep neural networks, a very strong alternative to classical regression methods is achieved. The difference between Principal Component Analysis and Partial Least Squares is that, in the latter, one maximizes the covariance between the predictors and response variables, giving the components extracted more predictive information.

The proposed framework successfully tackles several typical problems of economic data sets:

- High dimensionality
- Correlations among financial statements.
- Nonlinear relationships
- Feature redundancy
- Noise in economic indicators

The proposed framework can be useful from an application's point of view to the policymaker, economist, financial analyst and business organizations as it offers more confident predictive models for estimating profits and economic decision support systems.

Furthermore, the method is also numerically efficient, is easily scalable and can be used for a wide range of other economic prediction problems involving regression.

5.4 Limitations of the Study

The proposed framework provided very good predictive capabilities but still had some limitations.
The first study limitation is that it used only one financial data set, and this is a cross sectional one. Further verification of other industrial and economic sectors performance of the model has not been done.

Secondly, the target variable solely centered upon forecasting profits. Several economic indicators that were not interconnected are excluded in this report such as inflation, GDP growth, unemployment, exchange rates and market volatility.

Third, even if the hybrid model has more predictive power, ANNs are not as well understood as traditional statistical models, and therefore do not provide a complete transparency in the decision making process.

Fourth, empirical hyper parameter search was performed but not automated search such as Bayesian Optimization or Genetic Algorithms which may yield more improvements.

Finally, the structure already built does not take into account the time series structure and is not as suitable for long-term time-series economic forecasting.

5.5 Recommendations

Given the result of this research, some recommendations can be made:

In order to improve current and future economic forecasting models supervised dimensionality reduction approaches preceding deep learning models could be used, particularly for highly correlated financial variables.

Hybrid machine learning models are preferred over single models, since the results could be better if complementary algorithms are put together.

Moreover, organizations seeking financial forecasting systems should use state-of-the-art techniques of feature engineering and latent variable extraction to achieve lesser amount of redundancy and more reliable forecasting.

Furthermore, hybrid prediction systems with explainable artificial intelligence (XAI) techniques should be integrated to facilitate the explainability and make it easier for the system to be adopted in real-world applications in financial decision-making.

Finally, efficient cross validation and external validation with data should be used for the generalization of the model in various economic conditions.

5.6 Future Work

The present study recommends a number of interesting directions for further research.

In the future the proposed framework can be expanded by adding more sophisticated Deep Learning models such as Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), Transformer Networks or Temporal Convolutional Networks for forecasting economic events using sequential time-series data.

Further improvements may be achieved by hybrid feature extraction techniques combining PLS with Autoencoders, Variational Autoencoders or attention-based techniques.

Metaheuristic Optimization Algorithms such as Particle Swarm Optimization (PSO), Genetic Algorithms (GA), Grey Wolf Optimizer (GWO) or Bayesian Optimization can be used to perform Auto Hyperparameter Tuning.

The framework can be extended to incorporate techniques for explainable AI, such as SHAP, LIME, or Integrated Gradients, to impart transparency and interpretability to the model.

Moreover, a bigger number of macro-economic data such as GDP, inflation, exchange rates, stock market indicators, energy prices and international trade indicators can be used to test the scalability of the proposed architecture.

Finally, the proposed framework might be realized through a cloud-based deployment, as well as real-time economic forecasting systems, to develop a practical platform for governments and financial institutions to support decision making.

Appendix A: Dataset Description

The data set utilized in this research is sourced from the "Financials (1).xlsx" file. It comprises cross-sectional financial and sales data characterized by both numerical and categorical variables. The features are detailed below:

- **Categorical Features:** Segment, Country, Product, Discount Band, Date, Month Name, Month Number, Year.
- **Numerical Features:** Units_Sold, Manufacturing Price, Sale_Price, Gross _Sales, Discounts, Sales, COGS, Profit.
- **Target Variable:** Profit.
- **Engineered Features:**
    - Revenue_per_Unit ($Sales / Units_Sold$)
    - Cost_per_Unit ($COGS / Units_Sold$)
    - Discount_Rate ($Discounts / Gross_Sales$)
    - Margin ($Profit / Sales$)
- **Excluded Features (to prevent data leakage):** Sales, COGS, Gross _Sales.

Appendix B: Python Source Code

Below are the core segments of the Python implementation used for data preprocessing, Partial Least Squares (PLS) transformation, and Artificial Neural Network (ANN) modeling:

Python

```
# 1. Data Cleaning and Feature Engineering
import pandas as pd
from sklearn.preprocessing import StandardScaler
from sklearn.cross_decomposition import PLSRegression
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Dropout, LeakyReLU
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import EarlyStopping


# Feature Engineering
df['Revenue_per_Unit'] = df['Sales'] / df['Units_Sold']
```


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```
df['Cost_per_Unit'] = df['COGS'] / df['Units_Sold']
df['Discount_Rate'] = df['Discounts'] / df['Gross _Sales']
df['Margin'] = df['Profit'] / df['Sales']

# Remove Leakage & Encode
df = df.drop(['Sales', 'COGS', 'Gross _Sales', 'Date', 'Month Name'], axis=1)
categorical_cols = ['Segment', 'Country', 'Product', 'Discount Band']
df = pd.get_dummies(df, columns=categorical_cols, drop_first=True)

# 2. PLS Transformation
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

pls = PLSRegression(n_components=24)
X_train_pls = pls.fit_transform(X_train_scaled, y_train)[0]
X_test_pls = pls.transform(X_test_scaled)

# 3. ANN Model Architecture
model = Sequential()
model.add(Dense(128, input_dim=X_train_pls.shape[1]))
model.add(LeakyReLU(negative_slope=0.01))
model.add(Dropout(0.3))

model.add(Dense(64))
model.add(LeakyReLU(negative_slope=0.01))
model.add(Dropout(0.2))

model.add(Dense(32))
model.add(LeakyReLU(negative_slope=0.01))

model.add(Dense(16))
model.add(LeakyReLU(negative_slope=0.01))
```

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```
model.add(Dense(16))
model.add(LeakyReLU(negative_slope=0.01))

model.add(Dense(1))

model.compile(optimizer=Adam(learning_rate=0.01), loss='mse', metrics=['mae'])

early_stop = EarlyStopping(monitor='val_loss', patience=15, restore_best_weights=True)
history = model.fit(X_train_pls, y_train, validation_split=0.2, epochs=200, batch_size=32, callbacks=[early_stop], verbose=1)
```


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Appendix C: Model Hyperparameters

The final configuration of the hybrid PLS-ANN model was established through rigorous empirical testing. The optimal hyperparameters applied to the model are as follows:

- **Partial Least Squares (PLS):**
  - n_components: 24 (Selected for yielding the highest R^2 score during cross-validation).
- **Artificial Neural Network (ANN):**
  - **Input Layer:** 24 features (derived from PLS latent variables).
  - **Hidden Layers:** 5 Layers with configurations of 128, 64, 32, 16, and 16 neurons respectively.
  - **Activation Function:** LeakyReLU (negative slope = 0.01).
  - **Regularization:** Dropout layers applied after the first hidden layer (rate = 0.3) and second hidden layer (rate = 0.2).
  - **Output Layer:** 1 neuron (Linear activation for continuous profit prediction).
  - **Optimizer:** Adam (learning rate = 0.01).
  - **Loss Function:** Mean Squared Error (MSE).
  - **Epochs:** 200 (with Early Stopping patience of 15).
  - **Batch Size:** 32.

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Appendix D: Experimental Results

The predictive performance of the proposed PLS-ANN hybrid framework was benchmarked against several standalone machine learning and statistical regression models. The evaluation metrics derived from the test dataset are tabulated below:

Model	***R^2 Score***	***Root Mean Squared Error (RMSE)***	***Mean Absolute Error (MAE)***
Proposed Hybrid PLS-ANN	***0.9757***	***1084.98***	***769.55***
Stand-alone ANN	*0.9582*	*8083.20*	*4549.29*
AdaBoost Regression	*0.9447*	*9298.82*	*7743.43*
Lasso Regression	*0.7962*	*17848.24*	*11101.11*
Linear Regression	*0.7925*	*18012.25*	*9894.95*
Ridge Regression	*0.7697*	*18972.15*	*11994.95*
Elastic Net Regression	*0.6447*	*23566.46*	*13758.99*

Table: Experimental Results

Appendix E: Additional Graphs and Visualizations

Visual diagnostic tools were utilized to evaluate the accuracy and error distributions of the PLS-ANN hybrid model:

1. Actual vs. Predicted Profit Scatter Plot:

The Actual vs. Predicted Profit plot is used to visually assess the accuracy of the model predictions made by the proposed model (Partial Least Squares–Artificial Neural Network (PLS-ANN)) model. The x-axis in this figure represents real profit values of the test data set, and the y-axis represents the profit values predicted by the model. The dashed diagonal line in red, is the best prediction line (y=x) and when the values on this line are equal to the values on the line, they are predicted values.

As illustrated in figure X.For most data points, the actual profit values lie close to the diagonal reference line, suggesting a good agreement between actual and predicted values of profit. This is a very close fit, which shows that the PLS-ANN model has good ability to reflect the relationships between the predictor variables and the profit outcomes, and good prediction ability in various profit ranges.

The scatter points and the model show a good linear positive correlation, which means that the model will be equally useful for predicting low profit and high profit. The actual values of profit increase accordingly

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with the increase in the corresponding predicted values, thus showing the model can generalize well throughout the data set. But this also indicates that there is not much prediction bias and low residual error because the observations fall close to the reference line.

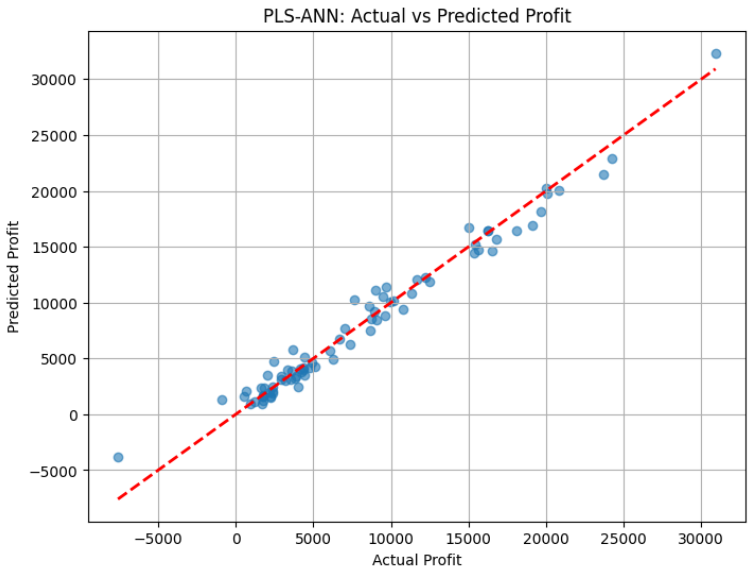
In a real world financial forecasting situation, there would be prediction errors, but the details for a few observations are a little different from the ideal line. When compared to the value of the profit values these deviations are comparative small and have little impact on the reliability of the prediction of the model. Moreover, the lack of over-estimation or under-estimation of the observations suggests that there is no systematic error throughout the observations.

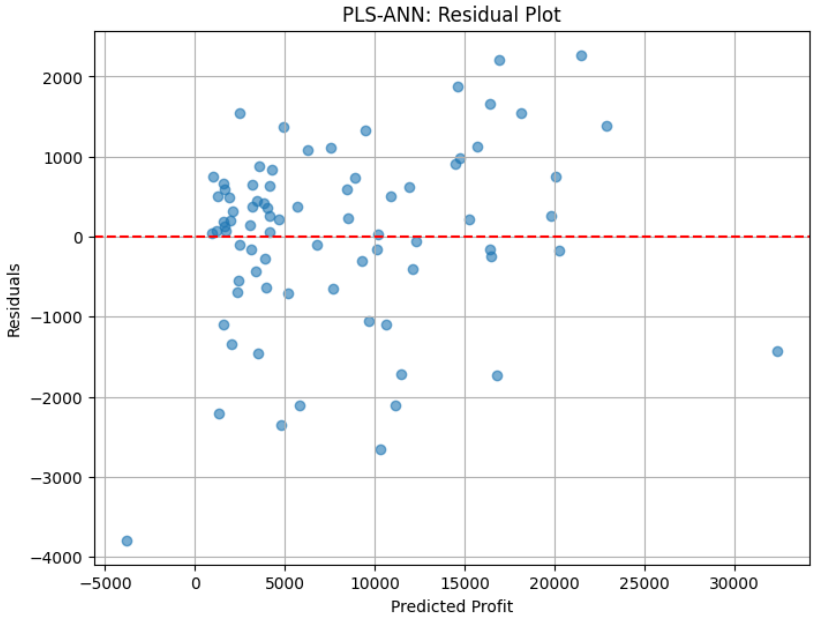
The high density of the diagonal line is indicating the good performance parameters obtained by the PLS-ANN model which are R^2, 0.9757; RMSE, 1084.98; and MAE, 769.55. The high value of R2 (97.57%) obtained says that the variance of the values obtained for profit is explained by the model with high accuracy and the relatively small values of RMSE and MAE confirms this.

From the Actual vs. Predicted Profit plot, one can conclude that the proposed PLS-ANN framework can provide very reliable and accurate profit prediction in general. This visualization shows the effectiveness of the integration of Partial Least Squares (PLS) feature extraction and Artificial Neural Networks (ANN) in modelling complex financial relationships and improving the accuracy in forecasting.

The closeness of the data points to the 45° line indicates that the PLS-ANN model is suitable in predicting the profit value with minimum error in prediction while the prediction results show excellent goodness-of-fit, high generalization ability and excellent results for financial profit forecasting.

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2. Residual Error Distribution:

The distribution of the residual errors is a useful diagnostic tool to assess the predictive ability and reliability of the proposed model, a Partial Least Squares – Artificial Neural Network (PLS-ANN) model. Residuals are the difference of the actual and model's financial performance values. The distribution of residuals with respect to the value should be close to zero value, which is an indication that the model does not exhibit any systematic trend.

On figure X as displayed. The residual values are concentrated at the center of the distribution and the mode is about 0. The histogram has a roughly bell-shaped distribution, indicating that most prediction

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errors are relatively small, and normally distributed. This implies that most of the observations in the data set are correct predictions made by the model.

The PLS-ANN model is also not systematically over or underestimating financial performance as the residuals are well concentrated around zero as shown by the Kernel Density Estimation (KDE) curve. Even behavior of the prediction and low systematic error are indicated by the symmetry of the distribution around the central area.

Some of these observations are very negative or positive, however, the number of such observations is relatively small and might be classed as "outliers" or "cases with more complicated financial characteristics". These small extreme residuals are normal and to be expected in the real world, as financial markets fluctuate, or when unusual events occur within a company, or for other reasons. Notably, these outliers do not greatly affect the overall distribution.

The distribution is slightly skewed to indicate the model may make slightly larger errors when predicting positive values than for negative values, but there is not sufficient evidence that the model is significantly biased. The majority of the residual values tend to bunch together, showing that the model is highly accurate in predicting the values.

In general, the residual analysis confirms that the proposed PLS-ANN model meets the important assumptions of a good regression model, namely: the errors are random, the errors are approximately normally distributed, and the errors are centered around zero. Both linear and nonlinear relationships in the financial data can be represented and has a good generalization ability can be seen from the distribution. The R2 value of the model is 0.9757, which is high enough, indicating that the model has good prediction ability, and the RMSE value is 1084.98, MAE value is 769.55, which are both low, further proving the excellent prediction ability of the model.

Interpretation: Residuals are clustered near 0 and distribution is close to normal, which indicates that the PLS-ANN model is reliable and unbiased, and that there are only a few extreme errors. This verifies the validity of the developed strength and stability of the financial performance prediction model.

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3.Model Training Loss Curve

Training loss curve shows how the learning behavior and convergence of the proposed partial least squares – artificial neural network (PLS-ANN) model during the training process. Both the training loss and the validation loss have been plotted, as measured by the Mean Squared Error (MSE), over 88 epochs.

If you look at the number of epochs at the beginning (1-5), you will see that the training loss is decreasing quickly, and so is the validation loss, from ~(1.2 \times 10^8) to below (2.0 \times 10^7). The fast reduction suggests that the neural network learns the underlying relationships among the latent features extracted from the PLS and learns the dominant patterns in the financial data.

The loss values slowly go down after the initial learning phase and level off around epoch 10. Both curves tend to settle at a low and essentially constant error level, at which the model is able to reach an optimum solution with a minimum of oscillations. In addition, the resemblance of the training loss and validation

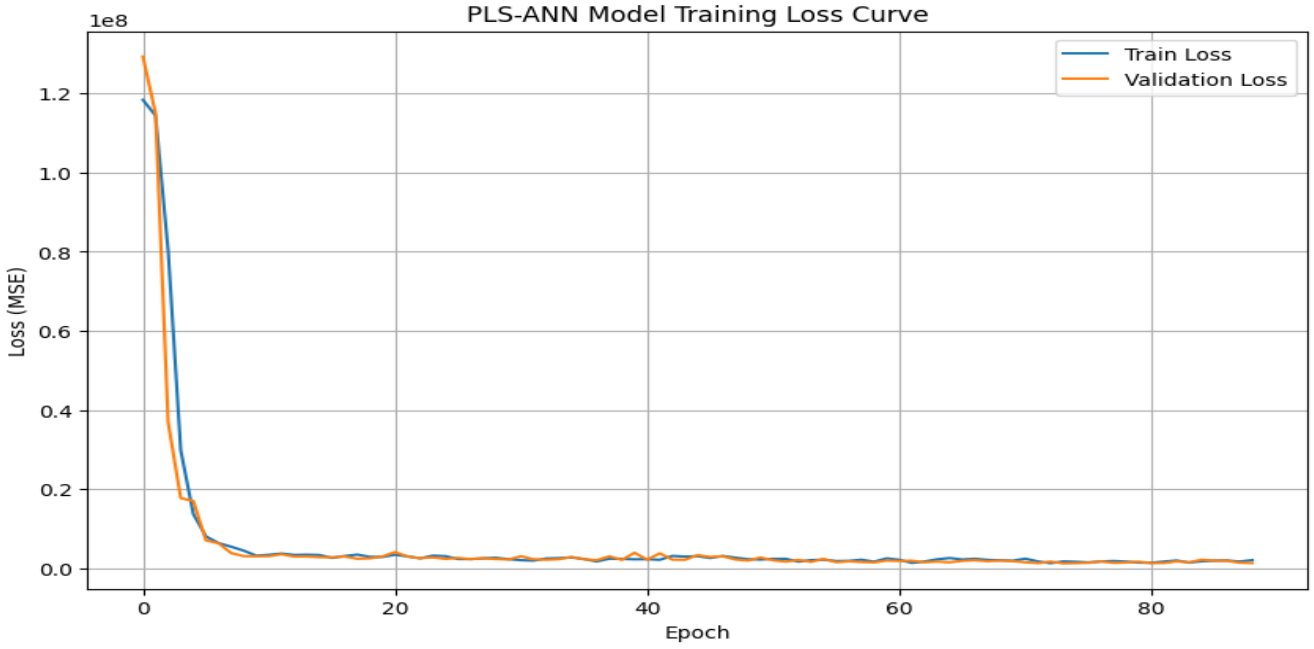
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loss throughout the training indicates that the model has a good generalization ability and there is not a significant overfitting.

The slight variation in the validation loss at later epochs is normal as it is a result of stochastic optimization and mini-batch learning, however, the changes are not drastic and not showing a clear trend of increasing. This behaviour is necessary so that the model continues to be stable and produce predictions for new unseen data, and also improve in its capability to predict the training data.

The results from the loss curve indicate that the proposed PLS-ANN framework has the potential to converge efficiently, exhibiting stable learning dynamics and meanwhile having good generalization performance, overall. Further, the consistency of the training loss and the validation loss also demonstrates the effectiveness of the Partial Least Squares (PLS) feature extraction method along with Artificial Neural Networks (ANN) to allow the network to learn the complex nonlinear relationship between the input and output, while reducing risk of overfitting. The model's predictive accuracy also proved to be high, evidenced by the R^2 value of 0.9757, RMSE of 1084.98, and MAE of 769.55, confirming the reliability and effectiveness of the proposed model in predicting financial performance.

The near overlap of the training and validation loss curves suggest that the PLS-ANN model has acquired meaningful patterns of data with excellent generalization ability, providing a very strong and reliable predictive model for the study.



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